RTL Design Sherpa

DDR2 / LPDDR2 Family Controller Hardware Architecture Specification 0.2

June 13, 2026

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2 FUB Breakdown

This section documents the **actual** Functional Unit Block decomposition as implemented in the rearchitected RTL: three integration layers under `pumice_core`, each built from leaf FUBs, driven by a PeakRDL CSR block in `pumice_top`.

2.1 Component Layout

Per the standard component layout:

```
pumice/
├── docs/                       # this HAS, MAS, generated PDFs
├── dv/                         # cocotb tests, BFM glue, tbclasses
└── rtl/
    ├── top/                    # pumice_top, pumice_core,
pumice_top_geared
    ├── macro/                  # the three layer macros +
pumice_csr.rdl + generated/
    ├── fub/                    # leaf FUBs (this section)
    ├── includes/               # shared `defines, package files
(pumice_pkg)
    └── filelists/
        ├── top/                # one .f per top
        ├── macro/              # one .f per layer
        └── fub/                # one .f per FUB
```

Each FUB has one top SystemVerilog file named `<fub>.sv` with module `<fub>`, a filelist under `filelists/fub/`, and a cocotb testbench under `dv/tests/`.

2.2 Layer Hierarchy

`pumice_top` instantiates the PeakRDL `pumice_csr` block (config by name) and `pumice_core`, which wires the three layers:

```
pumice_top                                  (CSR block + core)
└── pumice_core
    ├── pumice_axi4_ifc                          ("host AXI4 -> CAM-
buffered requests")
    │   ├── pumice_wr_intake                     AXI4 slave wr + AW-meta
FIFO + wr-data FIFO
    │   ├── pumice_rd_intake                     AXI4 slave rd + snarf
probe
    │   ├── addr_mapper                          flat AXI addr -> (rank,
bank, row, col)
    │   ├── pumice_wr_data_cam                   WR CAM + wr-data SRAM
(fill/commit-drain/snarf)
    │   └── pumice_rd_cmd_cam                    RD CAM + rd SRAM
(return-fill/drain)
    ├── pumice_mem_cmd_scheduler                 ("what command to issue
this cycle")
    │   ├── pumice_cmd_arbiter                   single pick core (open-
page inline)
    │   ├── pumice_bank_timers (bank_timer)      per-(rank,bank) FSM-
free JEDEC "safe" timers
    │   ├── global_timers                        tFAW / tRRD / tWTR /
tRTW / tCCD turnaround
    │   ├── refresh_ctrl                         tREFI postponer,
REFab/REFpb dispatch
    │   ├── init_sequencer                       DDR2 + LPDDR2 JEDEC MR
init
    │   └── mode_register                        MR shadow -> live
CL/CWL/BL/AL
    └── pumice_dfi_layer                         ("single CDC + DFI v2.1
datapath")
        ├── pumice_dfi_cdc                       the ONE clock crossing
(async gaxi FIFOs)
        ├── pumice_dfi_cmd_path                  unpack cmd, drive DFI
via dfi_cmd_formatter
        │       (uses dfi_cmd_formatter + dfi_signal_pack)
        ├── pumice_dfi_wr_serializer             commit-drain WR CAM ->
dfi_wrdata + mask
        └── pumice_dfi_rd_aligner                dfi_rddata capture ->
return-fill RD CAM
```

Also present but not in the default top build: `page_predictor` and `powerdown_ctrl` (referenced / optional; verify against the filelists).

2.3 FUB Inventory

2.3.1 pumice_axi4_ifc

2.3.1.1 pumice_wr_intake

- **Purpose**: AXI4 slave write engine. AW/W/B handshakes with an AW-meta FIFO and a wr-data FIFO; splits each host burst at DRAM-burst byte boundaries so each command maps to one DRAM burst.
- **Key params**: `AXI_DATA_WIDTH`, `AXI_ID_WIDTH`, `AXI_ADDR_WIDTH`, `BL`, `NUM_ENTRIES`.
- **Downstream**: `addr_mapper`, `pumice_wr_data_cam`.

2.3.1.2 pumice_rd_intake

- **Purpose**: AXI4 slave read engine; splits the read burst; probes the write-data CAM for a read-your-write snarf hit (unscheduled, same-id, same-BL) before committing the read command.
- **Key params**: `AXI_DATA_WIDTH`, `AXI_ID_WIDTH`, `AXI_ADDR_WIDTH`, `BL`, `NUM_ENTRIES`.

2.3.1.3 addr_mapper

- **Purpose**: Decode a flat AXI address into (rank, bank, row, col) using the single `bank_lsb` knob: the bank field slides within the byte-offset-stripped word address and the column auto-splits below (`col_lo`) and above (`col_hi`) it, with the row/rank positions invariant. An optional bank XOR-hash folds row bits (+ a seed) into the bank index. Combinational, single stage.
- **Key params**: `AXI_ADDR_WIDTH`, `NUM_RANKS`, `NUM_BANKS`, `ROW_WIDTH`, `COL_WIDTH`.
- **Runtime inputs**: `bank_lsb_i`, `hash_en_i`, `hash_seed_i` (`ADDR_MAP.bank_lsb` / `hash_en` / `hash_seed`).
- **Notable**: Mirrors the Python address-mapping class in the DV repo; the same decode drives RTL and BFM. There is no scheme selector.

2.3.1.4 pumice_wr_data_cam

- **Purpose**: Write-data CAM. Fills each write burst into an SRAM and records the command; provides scheduler lookup / oldest / commit ports; commit-drains the burst to the DFI write serializer; and is the snarf source for read

forwarding. An `r_fdone` fill-complete flag gates schedulability and snarf. Streaming read engine is FIFO-fed / oldest-pick — no active-slot state latch.

- **Key params**: `NUM_ENTRIES`, `N_SRAM_SLOTS`, `NUM_RANKS`, `NUM_BANKS`, `ROW_WIDTH`, `BL`.

2.3.1.5 pumice_rd_cmd_cam

- **Purpose**: Read-command CAM / reorder buffer. Records read commands; return-fills DRAM read beats into an SRAM; oldest-first drain engine (gated on data-ready) streams beats back to `pumice_rd_intake`. No active-slot latch.
- **Key params**: `NUM_ENTRIES`, `N_SRAM_SLOTS`, `NUM_RANKS`, `NUM_BANKS`, `ROW_WIDTH`, `BL`.

2.3.2 pumice_mem_cmd_scheduler

2.3.2.1 pumice_cmd_arbiter

- **Purpose**: The single command-pick core. Picks one abstract command per cycle from {ACT, RD/RDA, WR/WRA, PRE, REF, MRS, NOP} from the CAM lookups, gated by the per-(rank,bank) `safe_*` from `pumice_bank_timers` and the turnaround windows from `global_timers`. The open-page decision is inline.
- **Page policy**: OPEN (row-hit reuse + explicit PRE on miss), CLOSE (auto-pre on every column op via RDA/WRA), or HAPPY_HYBRID (predictor-selected). Programmed via `REFRESH_TUNING.page_policy_or`.
- **Key params**: `NUM_ENTRIES`, `NUM_RANKS`, `NUM_BANKS`, `AGE_WIDTH`.
- **Notable**: A combinational picker with registered feedback — always macro-test it, since registered-feedback latency created double-issue hazards caught only at the layer level.

2.3.2.2 pumice_bank_timers (instantiates bank_timer)

- **Purpose**: Stamps one `bank_timer` per (rank, bank) and fans the arbiter's command-event strobes to the addressed instance. `bank_timer` is FSM-free: preset/decrement countdown timers (tRCD, tRAS, tRC, tRP, precharge-block) + a row-open register + a single auto-precharge bit. The per-command `safe_*` outputs are combinational off the timers (one register stage), so the arbiter sees the just-issued command's effect one cycle later with no multi-stage lag.
- **Key params**: `NUM_RANKS`, `NUM_BANKS`, `ROW_WIDTH`.

- **Runtime inputs**: `t_rcd_i`, `t_rp_i`, `t_ras_i`, `t_rc_i`, `t_wr_i`, `t_rtp_i`.

2.3.2.3 `global_timers`

- **Purpose**: Cross-bank / cross-rank turnaround windows: tFAW (4-ACT window), tRRD, tWTR, tRTW, tCCD. ANDed by the arbiter.
- **Runtime inputs**: `t_faw_i`, `t_rrd_i`, `t_wtr_i`, `t_rtw_i`, `t_ccd_i`.

2.3.2.4 `refresh_ctrl`

- **Purpose**: tREFI down-counter; postponed-refresh accumulator (JEDEC max 8); refresh-pending signal to the arbiter; REFab / REFpb dispatch (`REFRESH_TUNING.refpb_policy_or`).
- **Runtime inputs**: `t_refi_i`, `refresh_burst_i`.

2.3.2.5 `init_sequencer`

- **Purpose**: Cold-boot engine that runs the memtype-specific JEDEC MR/init sequence (DDR2 and LPDDR2), driving CKE / RESET_N and MR-write strobes into `mode_register`, and holds off traffic until init completes.
- **Runtime inputs**: `t_init_wait_i`, `t_dll_wait_i`, `t_mrd_wait_i`, `t_rp_wait_i`, `t_rfc_wait_i`, `memtype_i`.

2.3.2.6 `mode_register`

- **Purpose**: Per-rank Mode Register shadow + live decode to CL / CWL / BL / AL (memtype-dependent), plus drive-strength and ODT-rule outputs consumed by the DFI layer.
- **Key params**: `NUM_RANKS`, `MAX_MR_IDX` (17; covers DDR2 MR0..3 and LPDDR2 MR0..16).

2.3.3 `pumice_dfi_layer`

2.3.3.1 `pumice_dfi_cdc`

- **Purpose**: The single controller-to-PHY clock crossing, built from asynchronous gaxi FIFOs (command, write-data, read-data). One FIFO word is one DFI cycle, so the datapaths are bubble-free.
- **Key params**: `CMD_FIFO_DEPTH`, `WD_FIFO_DEPTH`, `RD_FIFO_DEPTH`, `N_FLOP_CROSS`.

2.3.3.2 `pumice_dfi_cmd_path` (uses `dfi_cmd_formatter` + `dfi_signal_pack`)

- **Purpose**: DFI-domain command path. Pops the abstract command stream from the CDC FIFO, unpacks {op, rank, bank, row, col, ap}, and drives the

multi-phase DFI command bus via `dfi_cmd_formatter`; emits wr_fire / rd_fire strobes so the serializer / aligner can schedule their data phases.

- **`dfi_cmd_formatter`**: JEDEC command encoding into DFI wires — DDR2 drives cs_n / ras_n / cas_n / we_n / address / bank (including the A10 auto-precharge bit and ODT rule); LPDDR2 packs the 10-bit CA-bus command (two edges) onto `dfi_address`, per a `memtype` branch. Swap this when moving DFI generations.
- **`dfi_signal_pack`**: multi-phase aggregation — widens each DFI control bus to per-phase × `DFI_RATE`.

2.3.3.3 `pumice_dfi_wr_serializer`

- **Purpose**: On wr_fire, commit-drains the write burst from `pumice_wr_data_cam`'s SRAM and drives `dfi_wrdata` / `dfi_wrdata_en` / `dfi_wrdata_mask` with PHY alignment.
- **AXI ↔ DFI mask polarity**: AXI `wstrb`=1 means write; DFI `mask`=1 means do-not-write → `dfi_wrdata_mask = ~wstrb`.

2.3.3.4 `pumice_dfi_rd_aligner`

- **Purpose**: Drives `dfi_rddata_en` `t_rddata_en` cycles after a READ command; captures `dfi_rddata` beats; return-fills them into `pumice_rd_cmd_cam`.
- **Runtime inputs**: `t_rddata_en_i`, `t_phy_wrlat_i`, `rd_phase_i`, `wr_phase_i`.

2.4 Data Width and Gearing

Two distinct concepts:

1. **Internal DFI geometry.** `DW = DRAM_BEAT_WIDTH * DFI_RATE` (128 default). One AXI beat is one DFI word is `DFI_RATE` DRAM beats; one AXI burst is one DRAM burst. The DW → per-phase split happens in `dfi_signal_pack` / the DFI layer. The host AXI data width equals `DW` at `pumice_top`.
2. **Host-width freedom.** The optional `pumice_top_geared` wrapper adds a free `HOST_AXI_DATA_WIDTH` and inserts the repository's formally-verified `axi4_dwidth_converter_wr` / `_rd` between a host-width slave and the fixed-DW core. `HOST_AXI_DATA_WIDTH == DW` is a bit-identical generate bypass. Verified host ∈ {64, 128, 256}.

2.5 What Changed vs the Earlier Architecture

The controller was rearchitected from an earlier FSM-based decomposition. The following blocks were retired; their behavior now lives in the FUBs shown:

Retired block	Replaced by
`txn_queue`	the two CAMs: `pumice_wr_data_cam` + `pumice_rd_cmd_cam`
`bank_machine`	FSM-free `bank_timer`, stamped by `pumice_bank_timers`
`xbank_timers`	`global_timers` (turnaround) + `pumice_bank_timers` (per-bank)
`cmd_encoder`	`dfi_cmd_formatter` (+ `dfi_signal_pack`)
`odt_ctrl`	ODT inside `dfi_cmd_formatter` / `mode_register` (no standalone block)
`page_predictor` (standalone)	open-page decision inline in `pumice_cmd_arbiter` (`page_predictor.sv` optional)
`wr_cmd_cam`	`pumice_wr_data_cam`
`scheduler`	`pumice_mem_cmd_scheduler` (`pumice_cmd_arbiter` + timers + refresh + init)
`gear_dfi`	`pumice_dfi_layer`
`axi_frontend` / `axi_intake` / `*_macro-as-architecture`	`pumice_axi4_ifc`, generated `pumice_csr`, `pumice_core`

The old names may still appear in retired sentinel tests; they are not part of the current architecture.

2.6 Integration

The three layer macros are structural wiring; behavioral logic lives in the leaf FUBs. The principal wiring concerns:

1. **Arbiter ↔ timing fan-out**: `pumice_bank_timers` exposes per-(rank,bank) `safe_*` off single-stage timers; `global_timers` exposes turnaround-OK signals. `pumice_cmd_arbiter` reduces these into one command per cycle.
2. **CAM ↔ DFI-datapath coupling**: `pumice_wr_data_cam`'s commit-drain feeds `pumice_dfi_wr_serializer`; `pumice_dfi_rd_aligner`'s return-fill advances `pumice_rd_cmd_cam`.

3. **The single CDC**: everything up to `pumice_dfi_cdc` is on `aclk`; the command path, serializer, and aligner are on `dfi_clk`.

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3 Document Information

3.1 DDR2 / LPDDR2 Family Controller Hardware Architecture Specification

Document Number: DDR2-LPDDR2-HAS-001 **Version:** 0.4 **Status:** Draft - reconciled with rearchitected RTL **Classification:** Open Source - Apache 2.0 License

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3.2 Document Purpose

This Hardware Architecture Specification (HAS) provides a high-level architectural overview of the DDR2 / LPDDR2 unified memory controller family. It describes the system-level design, external interfaces, characterization parameters, and integration requirements without detailing internal implementation specifics.

Target Audience:

- System architects evaluating the controller for SoC integration
- Hardware engineers planning RTL implementation
- Verification engineers planning system-level testing
- Software engineers developing low-level memory configuration

Companion Documents:

- DDR2/LPDDR2 Family Controller Pre-Architecture Spec (`../pre-aspec.md`) — bullet-form architecture rationale
- DDR2/LPDDR2 Paging and Refresh Notes (`../paging-refresh-notes.md`) — research-derived parameter sources

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3.3 Revision History

Version	Date	Author	Notes
0.1	2026-06-13	RTL Design Sherpa	First sketch — module decomposition
0.2	2026-06-14	RTL Design Sherpa	Top-level interface port list (§2.4), FUB SWAG (§2.5), family-wide config-bit registry (§5.4), multi-rank support across §2 / §3.3 / §3.4 / §3.6 / §5.2 / §6.3
0.3	2026-07-07	RTL Design Sherpa	Narrow-device (x16) support: distinguish DRAM beat vs physical device word (`DRAM_DEVICE_WIDTH`); device-word column granularity + burst-length scaling; `DFI_PHASE` (rd/wr phase) CSR. Fixes on-silicon DDR2 read failure (Nexys A7 x16 bring-up).
0.4	2026-07-12	RTL Design Sherpa	Full reconciliation with the rearchitected RTL: three-layer core (`pumice_axi4_ifc` / `pumice_mem_cmd_scheduler` / `pumice_dfi_layer`); FSM-free `bank_timer`; de-FSM'd CAMs; single `ADDR_MAP.bank_lsb` address knob (scheme selector retired); fully-functional LPDDR2 (bit-exact JESD209-2F CA + JEDEC MR init); host-width `pumice_top_geared` gearing; block/hierarchy diagrams regenerated.

3.4 Scope of This Document

This HAS describes a single parameterized memory controller covering DDR2 and LPDDR2. It defines:

- External interfaces (AXI4 slave, DFI v2.1 master, APB CSR slave)
- Module-level decomposition and behavioral specification
- Build-time and run-time configuration parameters
- Initialization, power-state, and refresh policies
- Verification strategy and characterization plan

It does not define:

- Bit-level pin assignments (the SystemVerilog port list in `regs/pumice_ctrl_ports.svh` is the canonical wire-level source)
- Detailed timing diagrams (deferred to v0.3)
- Verilog package skeletons or RTL stubs
- Floorplan or layout guidance

3.5 Out of Scope

The following features are explicitly out of scope for this version of the controller and will be addressed in higher-generation family controllers or future revisions:

- Inline ECC
- DFI training, frequency-change, and low-power sub-interfaces
- AXI exclusive access semantics
- Bank groups (not present in DDR2 / LPDDR2)
- Command/Address parity, CRC, Data Bus Inversion (not present in DDR2 / LPDDR2)

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4 Purpose and Scope

4.1 Document Purpose

This Hardware Architecture Specification (HAS) defines the high-level architecture of the DDR2 / LPDDR2 Family Memory Controller. It serves as the primary reference for:

- **System Integration** — understanding the controller's external interfaces and system-level requirements
- **RTL Implementation Planning** — module decomposition, interface contracts, and parameter sweeps
- **Verification Planning** — system-level test scenarios and the characterization sweep matrix

- **Driver Development** — programming model and APB register interface

This document complements the upcoming Micro-Architecture Specification (MAS), which will provide detailed block-level implementation specifics, signal-level pinouts, and timing diagrams.

4.2 Scope

This HAS covers a single unified controller that supports both DDR2 and LPDDR2 memory. The memory type is a runtime CSR selection (`PHY_TIMING.memtype`: 0 = DDR2, 1 = LPDDR2). The controller is built as three layers under `pumice_core`:

- **`pumice_axi4_ifc`** — AXI4 slave host interface: dumb write/read intakes, an address mapper, a write-data CAM (write buffer + read-your-write snarf source), and a read-command CAM (read reorder buffer).
- **`pumice_mem_cmd_scheduler`** — the command-scheduling layer: a single command arbiter (`pumice_cmd_arbiter`, open-page decision inline), per-(rank,bank) FSM-free JEDEC "safe" timers (`pumice_bank_timers` / `bank_timer`), global turnaround timers (tFAW / tRRD / tWTR / tRTW / tCCD), a refresh controller, an init sequencer (DDR2 and LPDDR2 JEDEC MR init), and a mode-register shadow (CL / CWL / BL / AL decode).
- **`pumice_dfi_layer`** — a single controller-to-PHY clock crossing (`pumice_dfi_cdc`, async FIFOs) plus the DFI-domain command path (`dfi_cmd_formatter` / `dfi_signal_pack`), write serializer, and read aligner, presenting the DFI v2.1 pin bus.

Configuration (timings, phases, page policy, address map, memtype) is delivered by name from a PeakRDL-generated CSR register block (`pumice_csr`) instantiated in `pumice_top`. An optional outer wrapper, `pumice_top_geared`, gives the host a free AXI data width by inserting the repository's formally-verified `axi4_dwidth_converter_wr` / `_rd` between a host-width slave and the fixed-width core.

The differences between DDR2 and LPDDR2 are the command encoding (DDR2 ras/cas/we vs the LPDDR2 10-bit CA bus, both in `dfi_cmd_formatter`), the init MR sequence in `init_sequencer`, and a small number of mode-register decode differences. Both memtypes pass the full simulation suite; everything else is shared.

4.3 Design Philosophy

This controller is intended as a **characterization-first** design. Algorithmic choices that have meaningful research-backed alternatives — page policy, refresh policy, refresh deferral depth, address mapping — are exposed as CSR fields rather than hardcoded. The verification plan includes a benchmark sweep over representative AXI traffic patterns to pick defaults from data, not assumption.

The principal characterization knobs, all programmed by name through the CSR block, are:

- **Page policy** (`REFRESH_TUNING.page_policy_or`: OPEN / CLOSE / HAPPY_HYBRID) — where HAPPY_HYBRID is the address-bit-based page-closure predictor from Ghasempour et al. (2015). The open-page decision itself is inline in `pumice_cmd_arbiter`.
- **Refresh policy** (`REFRESH_TUNING.refpb_policy_or`: ROUND_ROBIN / OLDEST_FIRST / DARP) — where DARP is the dynamic access refresh parallelization scheme from Chang et al. (HPCA 2014).
- **Address map** (`ADDR_MAP.bank_lsb`, `hash_en`, `hash_seed`) — a single knob that slides the bank field within the word address, subsuming the classic ROW_MAJOR / BANK_INTERLEAVE schemes, with an optional bank XOR-hash.

These are described in detail in Chapter 5.

4.4 Audience

This document assumes the reader is familiar with:

- AXI4 protocol semantics (handshakes, burst types, ID-based ordering)
- DFI v2.1 specification (control / write-data / read-data sub-interfaces)
- JEDEC DDR2 (JESD79-2F) and LPDDR2 (JESD209-2F) command tables
- Basic memory-controller concepts (open vs closed page policy, refresh deadlines, bank state machines)

Readers new to these topics should first consult the companion `paging-refresh-notes.md` in the family working area for an entry-point bibliography.

5 Document Conventions

5.1 Typographic Conventions

Style	Meaning
`monospace`	Module names, parameter names, signal names, register names
bold	Defined terms on first introduction; important warnings
italic	Emphasis; placeholder values to be replaced
`NUM_RANKS`	Top-level build-time parameter (all caps)
`bank_timer`	RTL module name (lowercase with underscores)
`tRC`, `tRCD`	JEDEC timing parameters (as in JESD79-2 / JESD209-2)

5.2 Numbering Conventions

- **Decimal** numbers have no prefix: `16`, `256`.
- **Hexadecimal** numbers are prefixed with `0x`: `0x800`, `0xFF`.
- **Binary** literals use Verilog notation when needed: `8'b1011_0010`.
- Bit ranges use the convention `[MSB:LSB]`, e.g., `dfi_address[19:0]`.

5.3 Signal Direction Conventions

For DFI signals, direction is from the **memory controller's** perspective:

- **MC → PHY** — driven by the controller
- **PHY → MC** — driven by the PHY (sampled by controller)

For AXI signals, direction follows the standard AXI4 convention with the controller as the **slave**.

5.4 Module Diagram Conventions

ASCII block diagrams in this document use the following conventions:

All diagrams in this document are rendered from Mermaid source files in `assets/mermaid/`. The Mermaid source files (`.mmd`) are linked alongside each rendered image so the diagrams can be edited and re-rendered.

- **Rectangular boxes with sharp corners** denote RTL modules.

- **Subgraph rectangles** group modules by functional layer.
- **Solid arrows** show primary data or control flow.
- **Dashed arrows** show configuration or observation paths (typically from the CSR slave to other modules).
- **Trapezoidal nodes** denote external interface ports (AXI, DFI, APB).

5.5 Cross-Reference Conventions

- References to other chapters: “see Chapter 4”
- References to specific sections: “see §4.1” or “see Section 4.1”
- References to other documents: “see `pre-aspec.md`” (relative path)
- References to external standards: “per JESD79-2F §x.y”

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6 Definitions and Acronyms

6.1 Acronyms

Acronym	Expansion
AHB	Advanced High-performance Bus (ARM AMBA)
APB	Advanced Peripheral Bus (ARM AMBA)
AP	Auto-Precharge (DRAM column command modifier)
APD	Active Power Down
AXI	Advanced eXtensible Interface (ARM AMBA)
AxLEN	AXI burst length — the `AWLEN` / `ARLEN` signals (beats = `AxLEN`+1). NOT the DRAM BL: one AXI burst at core width is `(AxLEN+1)/DFI_RATE` DFI words and splits into fixed-BL DRAM commands. Use “AxLEN” whenever burst length refers to the AXI side, to avoid confusion with the JEDEC BL below.
BL	Burst Length — JEDEC/DRAM burst (MR0 device beats: BL4/BL8). Distinct from AxLEN.

Acronym	Expansion
CA	Command/Address (LPDDR2/3/4 multiplexed bus)
CDC	Clock Domain Crossing
CL	CAS Latency
CMD	Command
CSR	Control/Status Register
CWL	CAS Write Latency
DARP	Dynamic Access Refresh Parallelization
DBI	Data Bus Inversion
DDR	Double Data Rate
DDR2	Double Data Rate 2 (JESD79-2)
DFI	DDR PHY Interface
DPD	Deep Power Down (LPDDR2)
FSM	Finite State Machine
FR-FCFS	First-Ready, First-Come-First-Served
HAPPY	Hybrid Address-based Page Policy in DRAMs
HAS	Hardware Architecture Specification
LPDDR2	Low Power DDR 2 (JESD209-2)
MAS	Micro-Architecture Specification
MC	Memory Controller
MR / MRS	Mode Register / Mode Register Set command
EMR / EMRS	Extended Mode Register / Extended MRS
NOP	No Operation
ODT	On-Die Termination
OOO	Out-of-Order
PASR	Partial Array Self-Refresh (LPDDR2 feature)
PHY	Physical Layer (DRAM electrical interface)
PRE	Precharge command
PREA	Precharge All Banks

Acronym	Expansion
RD	Read column command
RDA	Read with Auto-Precharge
REF	Refresh command
REFab	All-Bank Refresh (LPDDR2)
REFpb	Per-Bank Refresh (LPDDR2)
SDRAM	Synchronous Dynamic RAM
SoC	System on Chip
SR	Self-Refresh
TCSR	Temperature-Compensated Self-Refresh (LPDDR2)
WR	Write column command
WRA	Write with Auto-Precharge
ZQ	Impedance Calibration (ZQ Calibration)

6.2 Defined Terms

Bank Timer — A per-bank JEDEC "safe" timing tracker (`bank_timer`). It is *not* a finite state machine: it is a set of preset/decrement countdown timers (tRCD, tRAS, tRC, tRP, precharge-block) plus a small open-row register and a single auto-precharge bit. The per-command `safe_*` outputs are combinational off the timers (one register stage). `pumice_bank_timers` stamps one instance per (rank, bank).

Characterization Knob — A build-time or runtime CSR field intentionally exposed for performance sweeping during system characterization. Page policy, refresh policy, and address map are the principal knobs in this controller.

Command Formatter — `dfi_cmd_formatter`: the module that translates the controller's abstract command record into DFI wire signals. A single module with a memtype branch — DDR2 drives ras/cas/we; LPDDR2 packs the 10-bit CA bus (two edges) onto `dfi_address`. `dfi_signal_pack` performs the final per-phase wire packing.

Command Arbiter — `pumice_cmd_arbiter`: the single command-pick core. It scans CAM readiness against the bank and global timers, applies the page policy (the open-page decision is inline here), and emits one abstract DRAM command per cycle.

FR-FCFS — First-Ready, First-Come-First-Served. The baseline DRAM scheduling policy: ready commands beat unready, row-hits beat row-misses, older entries beat younger on ties. Established by Rixner et al. (ISCA 2000).

Gearing (two senses) — (1) The internal DFI-rate split: one AXI/DFI word is spread across `DFI_RATE` DRAM beats and packed to per-phase buses in `dfi_signal_pack` / the DFI layer. (2) Host-width gearing: `pumice_top_geared` inserts formally-verified AXI data-width converters so the host AXI data width can differ from the core width.

Init Sequencer — `init_sequencer`: the cold-boot engine that issues the memtype-specific JEDEC MR/init sequence, driving MR loads and commands and holding off traffic until init completes.

Page Hit / Row Hit — An access whose target row matches the currently open row in the target bank. Avoids the tRP + tRCD penalty.

Page Conflict / Row Miss — An access whose target row differs from the currently open row in the target bank. Pays tRP + tRCD before the access can begin.

Page Policy — The strategy for when to close an open row. Programmed via `REFRESH_TUNING.page_policy_or` (OPEN / CLOSE / HAPPY_HYBRID); the decision is inline in `pumice_cmd_arbiter`.

Command CAMs — The two content-addressable buffers that hold in-flight requests between the AXI interface and the scheduler: `pumice_wr_data_cam` (write data buffer + snarf source) and `pumice_rd_cmd_cam` (read reorder buffer). Both store burst data in an SRAM with FIFO-fed / oldest-pick streaming read engines and no active-slot state latch.

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7 Scope and Goals

7.1 High-Level Goals

This controller provides a single parameterized RTL implementation supporting both DDR2 and LPDDR2 memory devices via DFI v2.1. Primary design goals:

1. **Unified engine, one command formatter with a memtype branch.** DDR2 and LPDDR2 differ only at the wire-encoding layer (ras/cas/we vs the 10-bit CA bus, both in `dfi_cmd_formatter`) and in the init MR sequence; the command arbiter, bank/global timers, refresh policy, init sequencer, address mapper, and AXI interface are shared between both targets. Memtype is a runtime CSR selection (`PHY_TIMING.memtype`).

2. **Embedded / mobile / mid-range SoC targets.** Not optimized for high-end server DRAM (DDR5, RDIMM, multi-rank). Optimized for power efficiency and area economy.
3. **Characterization-first parameterization.** Algorithmic choices with research-backed alternatives are exposed as build-time parameters so the design can be swept on real workloads before defaults are locked.
4. **Verification-friendly.** Mirrors the data structures and conventions of the DDR2 / LPDDR2 DFI BFM in the DV repository so the same address mapping, timing parameters, and command encodings can drive both RTL and simulation.

7.2 In Scope

The following features are explicitly within scope for this controller:

- **AXI4 slave** at the fixed core data width `DW = DRAM_BEAT_WIDTH * DFI_RATE`; an optional `pumice_top_geared` wrapper gives the host a free AXI data width (verified host ∈ {64, 128, 256}) via formally-verified data-width converters
- **DFI v2.1 master** driving the command, write-data, and read-data sub-interfaces
- **CSR register block** (PeakRDL-generated `pumice_csr`, passthrough cpuif) for configuration and observation, programmed by name
- **DDR2 and LPDDR2 device support** selected at runtime by the `PHY_TIMING.memtype` CSR field
- **DFI frequency ratio** of 1, 2, or 4 phases via the `DFI_RATE` parameter
- **Multi-rank operation** — 1, 2, or 4 ranks via `NUM_RANKS` parameter; per-rank `CS_n`, `CKE`, `ODT`, and refresh tracking; rank-aware ODT cross-termination rules per JEDEC
- **Page policies:** OPEN, CLOSE, HAPPY hybrid predictor (CSR-selected; open-page decision inline in `pumice_cmd_arbiter`)
- **Per-bank refresh** for LPDDR2 with DARP-style scheduling (parameterized)
- **All-bank refresh** for DDR2 (and as LPDDR2 fallback)
- **Elastic refresh deferral** up to JEDEC ceiling of 8× tREFI (parameterized)
- **Partial-array self-refresh (PASR)** for LPDDR2 with CSR-configurable mask
- **Temperature-compensated refresh** (LPDDR2 MR4 derating)

- **Cold-boot initialization** via microprogram step tables
- **Power state management** (Active, Active Power-Down, Self-Refresh, Deep Power Down)
- **Bring-up observability** via CSR-exposed counters (row-hit rate, refresh latency, queue depth)

7.3 Out of Scope

Explicitly **not** addressed in this controller version:

Feature	Reason
Inline ECC	Out-of-controller responsibility (SoC sideband)
DFI training sub-interface	Not required for v2.1 DDR2 / LPDDR2 operation
DFI frequency-change protocol	No use case for this generation
DFI low-power sub-interface	Power-state FSM uses CKE only
AXI exclusive accesses	Embedded targets don’t require
Bank groups	Not present in DDR2 / LPDDR2 (introduced in DDR4 / LPDDR4)
Command/Address parity	Not present in DDR2 / LPDDR2 (introduced in DDR4 / LPDDR4)
Write CRC	Not present in DDR2 / LPDDR2
Data Bus Inversion (DBI)	Not present in DDR2 / LPDDR2
DDR3 write leveling	Not present in DDR2 / LPDDR2 (LPDDR3 introduces)
DDR3 ZQCS / ZQCL on column	DDR2 uses OCD calibration; LPDDR2 uses MR10 ZQ
Higher DDR / LPDDR generations	Will be addressed in DDR3-LPDDR3 and DDR4-LPDDR4 controllers

7.4 Target Performance Envelope

Metric	Target
DRAM data rate	Up to DDR2-800 / LPDDR2-1066
AXI clock	Up to 200 MHz (typical embedded SoC)
Sustained read bandwidth	≥ 70% peak on stream workloads
99th-percentile read latency	≤ 200 ns (typical)
Refresh blocking overhead	< 5% (LPDDR2 with DARP); < 8% (DDR2)
Area	< 50K gates (excluding HAPPY predictor)

These are target envelopes for v1; not guarantees. They will be validated and refined during the characterization sweep.

7.5 Differentiating Features

This controller is informed by published memory-controller literature and the open-source memory-controller ecosystem (LiteDRAM in particular — used as the cross-validation reference; see §6.4), but it is not a re-implementation of any existing controller. The design's distinguishing features:

1. **Centralized FR-FCFS scheduler with full out-of-order issue.** A single scheduler scans the full transaction queue every cycle, considering all banks' readiness and row-hit state in one decision. This is closer to the commercial-vendor pattern (Synopsys, Cadence, Xilinx MIG) than to most open-source RTL controllers — open-source controllers typically expose OoO only at the cross-bank level (a per-bank FIFO model where bank parallelism is the only source of reordering). True per-port within-stream reordering — where a younger row-hit request bypasses an older row-conflict request — is uncommon outside vendor IP. AXI4 per-ID ordering is honored at the response side regardless. **For real-time / safety-critical / formally-verifiable systems**, OoO reordering can be disabled at runtime via the `SCHED_TUNING.force_inorder` CSR field (§3.2); the scheduler degrades to a first-ready FIFO that preserves arrival order at the cost of 10–25% sustained bandwidth on streaming workloads.

2. **Multi-rank support with JEDEC-compliant cross-rank ODT.** Most open-source DDR2/3 controllers are single-rank only — LiteDRAM's default config, UberDDR3, and most academic prototypes assume one rank per channel. This controller supports 1, 2, or 4 ranks via the `NUM_RANKS` build parameter. Per-rank `CS_n`, `CKE`, and `ODT` signals are driven; the rank-aware refresh manager tracks `last_ref_age` per (rank, bank); and the cross-rank ODT termination rules (read from rank N → ODT-high on rank ≠N during the read window; write to rank N → ODT-high on rank N during the write window) are handled

inside `dfi_cmd_formatter` / `mode_register` (no standalone ODT block) per JESD79-2 §x.y. Multi-rank is what makes the controller usable on real DIMM-class platforms rather than just on-board point-to-point DRAM.

3. **Unified DDR2 + LPDDR2 engine with one command formatter.** A single controller core supports both memtypes via the runtime `PHY_TIMING.memtype` CSR field. The LPDDR2 path includes the 10-bit CA bus encoding (JESD209-2F Table 60; packed two edges onto `dfi_address` in `dfi_cmd_formatter`), PASR mask handling (MR16/17), per-bank refresh (REFpb), temperature-compensated refresh (MR4 derating), and Deep Power Down — none of which are present in the open-source DDR2/3/4 reference controllers. LPDDR2 is fully functional (reads and writes) and passes the full simulation suite.

4. **DARP-style per-bank refresh scheduling** — picks the *idle* bank with the longest time since last refresh, falling back to oldest-first when no banks are idle. Based on Chang et al. (HPCA 2014); not present in available open-source controllers.

5. **Two-stage page management** — exact lookahead first, history-based fallback second.

    - Stage 1: a lookahead window (`SCHED_TUNING.lookahead_active`, 0 disables) provides exact auto-precharge decisions when the next same-bank request is already visible to the arbiter.
    - Stage 2: when lookahead is inconclusive (shallow queue or bursty traffic), the HAPPY hybrid page predictor (Ghasempour et al. 2015) uses address-bit-hashed saturating counters to predict the next access. Open-source controllers typically have either lookahead OR a fixed page policy. The two-stage combination is original to this design.

6. **Characterization-first parameterization.** Page policy, lookahead depth, refresh policy, refresh deferral, and address mapping are all runtime CSR fields with a defined sweep matrix (see §5 and §6.4). Most published controllers expose only a subset of these; few expose them all.

7. **Closed-page policy as an explicit option.** `REFRESH_TUNING.page_policy_or ∈ {OPEN, CLOSE, HAPPY_HYBRID}`. Most open-source controllers expose only on/off auto-precharge (effectively OPEN with optional lookahead). The CLOSE option is useful for adversarial / security-sensitive workloads where row-state side channels matter.

8. **Single-knob address mapping.** `ADDR_MAP.bank_lsb` slides the bank field within the byte-offset-stripped word address; the column auto-splits below (`col_lo`) and above (`col_hi`) it while the row/rank positions stay invariant. The classic schemes are just settings of this one knob (`bank_lsb == COL_WIDTH` = ROW_MAJOR; `bank_lsb == log2(cols/burst)` = maximum bank interleave), and an optional bank XOR-hash

(ADDR_MAP.hash_en / hash_seed) folds row bits into the bank index to defeat pathological row-conflict strides. There is no separate scheme selector.

9. **AXI4-first interface** with ID-aware out-of-order completion. The AXI integration is a first-class design constraint rather than a wrapper on a native protocol.

10. **Mirrored verification address-mapping object.** The addr_mapper mirrors a Python address-mapping class used by the DFI BFM, so the same mapping function drives both RTL and simulation. This eliminates a common source of address-decode bugs between RTL and verification.

7.5.1 Features Adopted from Standard Memory-Controller Patterns

These design patterns are widespread in the memory-controller literature and the open-source controllers we examined. Adopting them is the right call; they are noted here so the design rationale is transparent:

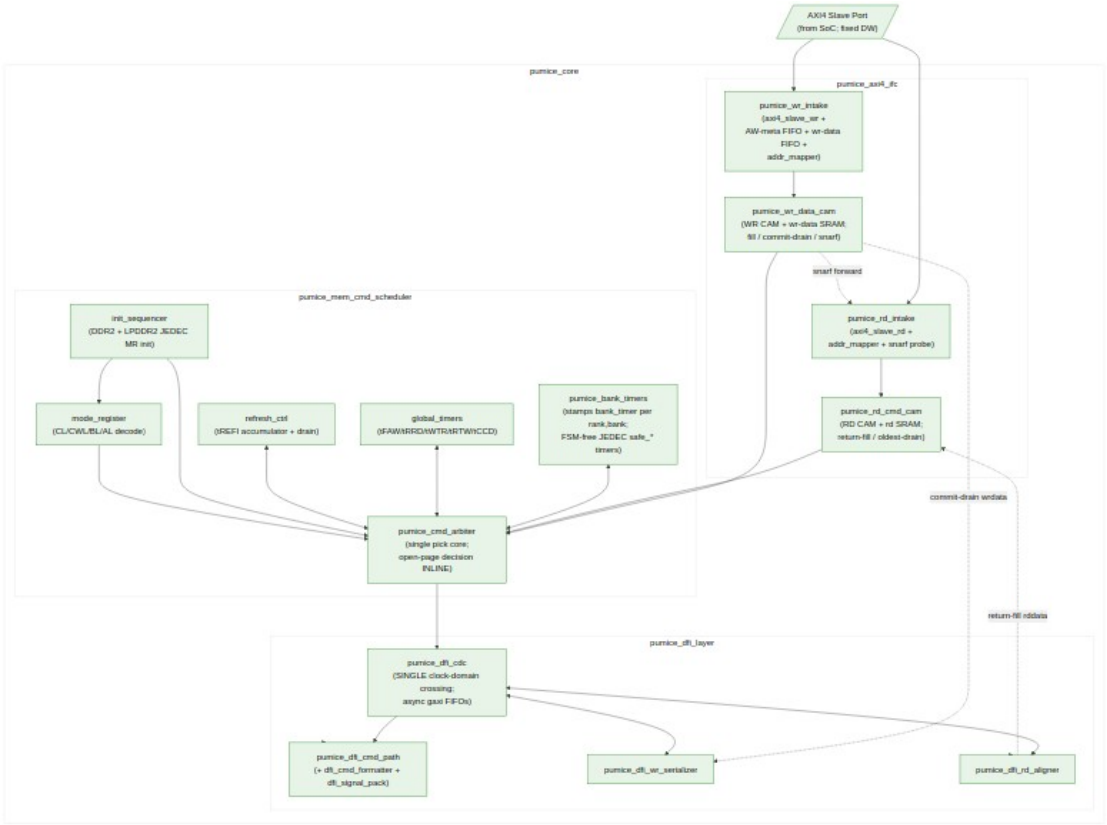
- **Per-bank FSM-free JEDEC "safe" timers** (bank_timer, stamped per (rank,bank) by pumice_bank_timers) + a central cross-bank/rank timer pool (global_timers: tRRD / tFAW / tWTR / tRTW / tCCD) — the standard bank/global split, here without a per-bank state machine.
- **FR-FCFS scheduler baseline** (Rixner et al. ISCA 2000) — the academic baseline; realized in pumice_cmd_arbiter.
- **Elastic Refresh deferral** (Stuecheli et al. MICRO 2010) — postponer with deterministic back-to-back drain, in refresh_ctrl.
- **Refresh arbitrated against bank readiness** — the arbiter waits for the addressed banks to be safe before issuing REF; established pattern.
- **Periodic ZQCS piggybacked on the refresh window** — established pattern for systems with thermal drift.
- **Init sequencer** issuing the memtype-specific JEDEC MR/init sequence (init_sequencer) — established pattern.

The design rationale for adopting versus extending each is documented in the relevant section.

8 Block Diagram

8.1 Top-Level Block Diagram

The controller's top-level data and control flow is shown below. AXI4 traffic enters at top-left into `pumice_axi4_ifc`, which maps addresses and pushes per-direction records into its two CAMs (`pumice_wr_data_cam`, `pumice_rd_cmd_cam`). The `pumice_mem_cmd_scheduler` layer queries the CAMs each cycle and picks the next abstract command (`pumice_cmd_arbiter` against the bank and global timers). The `pumice_dfi_layer` crosses the single controller-to-PHY clock boundary (`pumice_dfi_cdc`) and formats the chosen command into DFI v2.1 wires (`dfi_cmd_formatter` / `dfi_signal_pack`), while its write serializer and read aligner move write beats out and return read beats in alignment with the scheduled commands. `pumice_top` instantiates `pumice_core` (these three layers) plus the PeakRDL `pumice_csr` block that drives all configuration by name.



Top-Level Block Diagram

Source: 01_block_diagram.mmd

8.2 Data Flow Summary

Write path: 1. AXI master issues an AW/W transaction; `pumice_wr_intake` (an AXI4 slave write engine + AW-meta FIFO + wr-data FIFO) accepts it and splits the host burst at DRAM-burst boundaries. 2. `addr_mapper` translates the flat AXI address into (rank, bank, row, col) using `bank_lsb` (+ optional hash). 3. `pumice_wr_data_cam` fills the write burst into its SRAM and records the command; the `r_fdone` fill-complete flag gates schedulability and snarf. 4. `pumice_cmd_arbiter` queries the wr/rd CAMs against the per-(rank,bank) `safe_*` outputs from `pumice_bank_timers` and the turnaround windows from `global_timers`; it picks the next abstract command (ACT, WR/WRA, RD/RDA, PRE, REF, MRS, NOP) and applies the page policy (open-page decision inline). 5. In `pumice_dfi_layer`, the command crosses the CDC and `dfi_cmd_formatter` encodes it into DFI cs_n / ras_n / cas_n / we_n / address / bank wires (DDR2) or the packed CA bus (LPDDR2); closed-page uses WRA (auto-precharge). 6. `dfi_signal_pack` aggregates the per-phase DFI control bus. 7. On the write-fire strobe, the DFI write serializer commit-drains the burst from `pumice_wr_data_cam`'s SRAM and drives `dfi_wrdata` / `dfi_wrdata_en` / `dfi_wrdata_mask` with PHY alignment. 8. On retire, the write CAM slot is freed and `pumice_wr_intake` returns the B-response to the AXI master.

Read path: 1. AXI master issues an AR transaction; `pumice_rd_intake` accepts and splits the burst. 2. Address mapping identical to the write path; `pumice_rd_cmd_cam` records the read command. `pumice_rd_intake` probes the write-data CAM: on a snarf hit (unscheduled, same-id, same-BL) the read is served directly from the write CAM SRAM without going to DRAM (read-your-write forwarding). 3. `pumice_cmd_arbiter` issues ACT, then RD / RDA via the DFI layer. 4. The DFI read aligner drives `dfi_rddata_en` `t_rddata_en` cycles after the RD command and captures `dfi_rddata` beats for the burst. 5. Returned beats fill `pumice_rd_cmd_cam`'s SRAM; its oldest-first drain engine (gated on data-ready) streams them back out, tagged with the original AXI ID, on the R channel.

Refresh path: 1. `refresh_ctrl`'s tREFI counter elapses, incrementing the `refresh_pending` accumulator (JEDEC max 8 postponed). 2. When `refresh_pending` exceeds the soft threshold, refresh becomes highest-priority in the arbiter. 3. For REFab: the arbiter waits for the addressed banks to be safe (via `pumice_bank_timers`), then issues REF; the affected timers reload. 4. LPDDR2 per-bank refresh (REFpb) selection is driven by `REFRESH_TUNING.refpb_policy_or`.

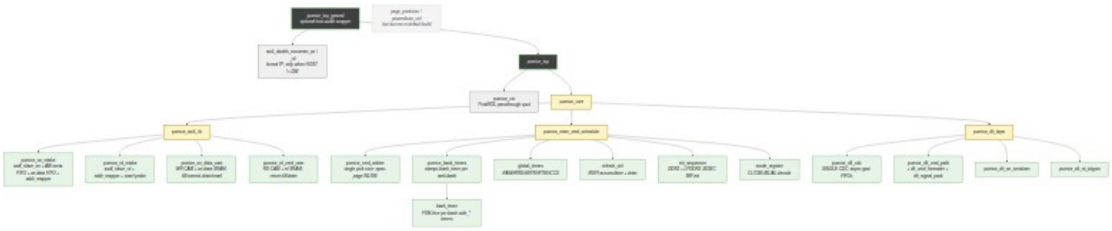
Init / power path: 1. On cold reset, `init_sequencer` holds off AXI traffic. 2. `init_sequencer` executes the memtype-specific JEDEC MR/init sequence, driving CKE / RESET_N and issuing MR-write strobes into `mode_register`. `mode_register` propagates live CL / CWL / BL / AL to the DFI layer. 3. On completion, `init_done_o` asserts and the arbiter begins servicing AXI traffic. 4. Power-down / self-refresh entry (Active / APD / SR / DPD) is managed by `powerdown_ctrl` (present but optional; not in the default top build), with SR-entry coordinated with `refresh_ctrl`.

Detailed per-module behavior follows in Chapter 3.

9 Module Hierarchy

The controller is decomposed into **three layers under `pumice_core`**, each built from leaf FUBs. Each FUB is independently verified at the unit level; each layer is verified as an integration ("macro") unit. See FUB Breakdown for the per-FUB role descriptions and Chapter 3 for the behavioral details.

9.1 Hierarchy Tree



Module Hierarchy

Source: 02_module_hierarchy.mmd

9.2 Layered Organization

```
SoC level
└── pumice_top_geared            ← OPTIONAL host-width wrapper
    │                              (axi4_dwidth_converter_wr/_rd when
HOST != DW)
    └── pumice_top               ← instantiated by the SoC
        ├── pumice_csr           ← PeakRDL passthrough cpuif register
block
        └── pumice_core          ← the controller proper
            ├── pumice_axi4_ifc          ← host AXI + wr/rd CAMs
            ├── pumice_mem_cmd_scheduler ← "what command to issue this
cycle"
            └── pumice_dfi_layer         ← single CDC + DFI v2.1
datapath
```

The single controller-to-PHY clock crossing lives inside `pumice_dfi_layer` (`pumice_dfi_cdc`, async FIFOs only). The host AXI interface, CAMs, and command scheduler all run on `aclk`; the DFI command path, write serializer, and read aligner run on `dfi_clk`.

9.3 Layer Groupings

Layer / macro	FUBs
pumice_axi4_ifc	pumice_wr_intake, pumice_rd_intake, addr_mapper, pumice_wr_data_cam, pumice_rd_cmd_cam
pumice_mem_cmd_scheduler	pumice_cmd_arbiter, pumice_bank_timers (bank_timer), global_timers, refresh_ctrl, init_sequencer, mode_register
pumice_dfi_layer	pumice_dfi_cdc, pumice_dfi_cmd_path (dfi_cmd_formatter, dfi_signal_pack), pumice_dfi_wr_serializer, pumice_dfi_rd_aligner
pumice_core	(wraps the three layers above)

Also present in the tree but not in the default top build: page_predictor and powerdown_ctrl (referenced / optional; verify against the filelists in rtl/filelists/).

Chapter 3 follows the layer ordering (one section per architectural layer, with module names hyperlinked to their MAS chapters).

9.4 Memtype-Conditional Logic

Memtype (DDR2 vs LPDDR2) is a runtime CSR selection (PHY_TIMING.memtype), not an elaboration parameter. The memtype-dependent logic lives in:

- **dfi_cmd_formatter** — a memtype branch: DDR2 drives ras/cas/we; LPDDR2 packs the 10-bit JESD209-2F CA-bus command (two edges) onto dfi_address.
- **init_sequencer** — runs the DDR2 or LPDDR2 JEDEC MR/init sequence.
- **mode_register** — CL / CWL / BL / AL decode differs by memtype.

Both memtypes pass the full simulation suite.

9.5 Single Clock-Domain Crossing

The design has exactly one clock-domain crossing: pumice_dfi_cdc inside pumice_dfi_layer, built from asynchronous gaxi FIFOs. One FIFO word is one DFI cycle, so the command / write-data / read-data datapaths are bubble-free. Everything up to the CDC is on aclk; everything past it is on dfi_clk.

9.6 What Changed vs the Earlier Architecture

The controller was rearchitected from an earlier FSM-based, elaboration-time decomposition into the current three-layer, CSR-driven design. Notably:

- **Retired**: `txn_queue`, `bank_machine`, `xbank_timers`, `cmd_encoder`, `odt_ctrl`, `scheme selector`, the `*_macro`-as-architecture names, and `axi_intake`/`axi_frontend`.
- **Replaced by**: the two CAMs (`pumice_wr_data_cam` + `pumice_rd_cmd_cam`), FSM-free `bank_timer` (stamped by `pumice_bank_timers`), `global_timers`, `dfi_cmd_formatter` (+ `dfi_signal_pack`), and the single `ADDR_MAP.bank_lsb` knob.
- **Introduced**: the single-CDC DFI layer, the PeakRDL `pumice_csr` block (configuration by name), and the optional `pumice_top_geared` host-width wrapper.

---

10 Top-Level Interfaces

This section is the integrator's quick-reference for the controller's top-level module port list. Detailed protocol semantics for each interface are in Chapter 4 (Interfaces); this is the **wire-level** view — names, directions, widths, parameter dependencies.

The top module is `pumice_top` and has four external interfaces:

1. AXI4 slave (host request path), signals prefixed `s_axi_*`
2. DFI v2.1 master (DRAM PHY path), signals suffixed `_o` / `_i`
3. CSR register cpuif (configuration and observation) — a PeakRDL passthrough bus, **not** APB
4. Clocks and resets (two domains) plus `init_done_o`

All four are exposed at the top level as flat per-signal ports. SystemVerilog interfaces are **not** used at the top boundary because the framework's interface convention (see CocoTBFramework BFM components) drives flat-port pin-level handshakes. Internal sub-blocks use interface objects, but the top-of-design is flat for synthesis and BFM compatibility.

An optional outer wrapper, pumice_top_geared, adds a HOST_AXI_DATA_WIDTH parameter and inserts formally-verified AXI data-width converters ahead of pumice_top; when HOST_AXI_DATA_WIDTH == DW the converters are bypassed and the build is bit-identical. This section describes pumice_top itself.

10.1 Parameter-Dependent Widths

The core data width is fixed by the DFI geometry: DW = DRAM_BEAT_WIDTH * DFI_RATE. There is no separate AXI_DATA_WIDTH parameter on pumice_top — the host AXI data width equals DW. Throughout this section:

- AW = AXI_ADDR_WIDTH (default 32)
- DFI_RATE = DFI phase count / frequency ratio (default 2)
- DRAM_BEAT_WIDTH = per-beat DRAM data width (default 64)
- DW = DRAM_BEAT_WIDTH * DFI_RATE (host AXI + DFI word width; default 128)
- SW = DW/8 (byte-strobe width)
- IW = AXI_ID_WIDTH (default 8)
- NR = NUM_RANKS (default 1)
- NB = NUM_BANKS (default 8)
- RW = ROW_WIDTH (default 14)
- CW = COL_WIDTH (default 10)
- BKW = $clog2(NUM_BANKS)
- CSR_ADDR_W = CSR address width (default 12)

10.2 1. AXI4 Slave (Host-side)

All host AXI ports carry the s_axi_ prefix.

10.2.1 Write Address Channel (AW)

Signal	Direction	Width	Description
s_axi_awid	input	IW	Write transaction ID
s_axi_awaddr	input	AW	Write start address
s_axi_awlen	input	8	Burst length – 1
s_axi_awsize	input	3	Beat size encoding
s_axi_awburst	input	2	Burst type (INCR mandatory; FIXED / WRAP optional)
s_axi_awlock	input	1	Lock (ignored — exclusives not

Signal	Direction	Width	Description
			supported)
s_axi_awcache	input	4	Cache hint (observed; no behavior)
s_axi_awprot	input	3	Protection bits (observed)
s_axi_awqos	input	4	QoS hint (boosts scheduler priority — see §3.2)
s_axi_awregion	input	4	Region (observed)
s_axi_awuser	input	1	User sideband (observed)
s_axi_awvalid	input	1	Address-valid
s_axi_awready	output	1	Address-ready

10.2.2 Write Data Channel (W)

Signal	Direction	Width	Description
s_axi_wdata	input	DW	Write data
s_axi_wstrb	input	SW	Byte enables
s_axi_wlast	input	1	Last beat of burst
s_axi_wuser	input	1	User sideband
s_axi_wvalid	input	1	Data-valid
s_axi_wready	output	1	Data-ready

10.2.3 Write Response Channel (B)

Signal	Direction	Width	Description
s_axi_bid	output	IW	Response ID
s_axi_bresp	output	2	Response code (OKAY / SLVERR)
s_axi_buser	output	1	User sideband
s_axi_bvalid	output	1	Response-valid
s_axi_bready	input	1	Response-ready

10.2.4 Read Address Channel (AR)

Signal	Direction	Width	Description
s_axi_arid	input	IW	Read transaction ID
s_axi_araddr	input	AW	Read start address

Signal	Direction	Width	Description
s_axi_arlen	input	8	Burst length – 1
s_axi_arsize	input	3	Beat size encoding
s_axi_arburst	input	2	Burst type
s_axi_arlock	input	1	Lock (ignored)
s_axi_arcache	input	4	Cache hint
s_axi_arprot	input	3	Protection
s_axi_arqos	input	4	QoS hint
s_axi_arregion	input	4	Region
s_axi_aruser	input	1	User sideband
s_axi_arvalid	input	1	Address-valid
s_axi_arready	output	1	Address-ready

10.2.5 Read Data Channel (R)

Signal	Direction	Width	Description
s_axi_rid	output	IW	Read data ID
s_axi_rdata	output	DW	Read data
s_axi_rresp	output	2	Response code
s_axi_rlast	output	1	Last beat
s_axi_ruser	output	1	User sideband
s_axi_rvalid	output	1	Data-valid
s_axi_rready	input	1	Data-ready

10.3 2. DFI v2.1 Master (PHY-side)

The DFI v2.1 sub-interfaces present on this controller are limited to **command**, **write-data**, **read-data**, and the **init handshake**. Training, frequency-change, low-power, update, and CRC sub-interfaces are not driven — see §2.1 Out of Scope.

The DFI ports are flat, per-phase-packed buses (the DFI_RATE phases are concatenated into one vector) rather than SystemVerilog arrays. All DFI ports carry the _o (output) or _i (input) suffix. Command formatting runs on dfi_clk, past the single CDC in pumice_dfi_layer.

10.3.1 DFI Command Bus

Signal	Direction	Width	Description
dfi_address_o	output	RW * DFI_RATE	Address operand (row / column); LPDDR2 CA-bus command is packed here
dfi_bank_o	output	BKW * DFI_RATE	Bank operand
dfi_cs_n_o	output	NR * DFI_RATE	Active-low chip-select, per rank per phase
dfi_ras_n_o	output	DFI_RATE	RAS strobe per phase (DDR2; idle for LPDDR2)
dfi_cas_n_o	output	DFI_RATE	CAS strobe per phase (DDR2; idle for LPDDR2)
dfi_we_n_o	output	DFI_RATE	WE strobe per phase (DDR2; idle for LPDDR2)
dfi_odt_o	output	NR * DFI_RATE	ODT, per rank per phase

10.3.2 DFI Write-Data Bus

Signal	Direction	Width	Description
dfi_wrdata_o	output	DW	Write data (all phases packed; = one DFI word)
dfi_wrdata_en_o	output	DFI_RATE	Per-phase write-data enable
dfi_wrdata_mask_o	output	SW	Per-byte write mask (DFI polarity: 1 = do-not-write)

10.3.3 DFI Read-Data Bus

Signal	Direction	Width	Description
dfi_rddata_en_o	output	DFI_RATE	Per-phase read-data enable
dfi_rddata_i	input	DW	Read data (all phases packed)
dfi_rddata_valid_i	input	DFI_RATE	Per-phase read-data valid

10.3.4 DFI Init Handshake

Signal	Direction	Width	Description
dfi_init_start_o	output	1	Init handshake — start of DRAM init
dfi_init_complete_i	input	1	PHY signals init complete

10.4 3. CSR Register Interface (Configuration / Observation)

Configuration is driven by name from a PeakRDL-generated register block (pumice_csr) via a passthrough **cpuif** request bus (not APB). Address width is CSR_ADDR_W (default 12). Detailed register map in §6.3.

Signal	Direction	Width	Description
s_cpuif_req	input	1	Request valid
s_cpuif_req_is_wr	input	1	1 = write, 0 = read
s_cpuif_addr	input	CSR_ADDR_W	Register address
s_cpuif_wr_data	input	32	Write data
s_cpuif_wr_biten	input	32	Per-bit write enable
s_cpuif_req_stall_wr	output	1	Write back-pressure
s_cpuif_req_stall_rd	output	1	Read back-pressure
s_cpuif_rd_ack	output	1	Read data valid
s_cpuif_rd_err	output	1	Read error
s_cpuif_rd_data	output	32	Read data
s_cpuif_wr_ack	output	1	Write accepted
s_cpuif_wr_err	output	1	Write error

10.5 4. Clocks, Resets, and Status

The design has two clock domains — the controller/AXI domain (aclk) and the DFI/PHY domain (dfi_clk) — with the single crossing inside pumice_dfi_layer.

Signal	Direction	Width	Description
aclk	input	1	Controller clock (host AXI, CAMs, scheduler)
aresetn	input	1	Controller reset, active low
dfi_clk	input	1	DFI/PHY clock

Signal	Direction	Width	Description
dfi_rstn	input	1	DFI reset, active low
init_done_o	output	1	Asserted when DRAM init completes

10.6 Hierarchical Top-Level View

The top-level pinout in shorthand:

```
pumice_top
    ├── AXI4 slave        (host)         s_axi_awid,
s_axi_awaddr, ..., s_axi_rvalid, s_axi_rready, ...
    ├── DFI v2.1 master   (PHY)          dfi_address_o, dfi_bank_o,
dfi_cs_n_o, ..., dfi_rddata_i, ...
    ├── CSR cpuif         (config)       s_cpuif_req,
s_cpuif_addr, ..., s_cpuif_rd_data, ...
    └── Clocks + status                  aclk, aresetn, dfi_clk,
dfi_rstn, init_done_o
```

The canonical SystemVerilog port list is `rtl/top/pumice_top.sv`. This section is the **integrator's overview**, not the build-time canonical list.

---

11 AXI4 Frontend

The AXI frontend is `pumice_axi4_ifc` (`rtl/macro/pumice_axi4_ifc.sv`). It bolts the repository's common AXI burst splitters onto two "dumb" 1:1 intakes and holds the two CAMs that decouple AXI-side handshake from the DRAM-side command scheduler:

```
host AXI4 -> [axi_master_wr_splitter] -> pumice_wr_intake ->
pumice_wr_data_cam
          -> [axi_master_rd_splitter] -> pumice_rd_intake ->
pumice_rd_cmd_cam
```

The intakes each contain a common `axi4_slave` (write or read side) plus an `addr_mapper` that translates the flat AXI address into `(rank, bank, row, col)`. The write intake also carries an AW-meta FIFO and a write-data FIFO; the read intake carries a snarf (read-your-write) probe into the write CAM. This chapter covers the splitters, the intakes, and `addr_mapper`. The two CAMs are covered in the data-paths chapter (`07_data_paths.md`).

11.1 Burst Splitters

`axi_master_wr_splitter` and `axi_master_rd_splitter` are formally-verified common modules. Each host burst is split at DRAM-burst-byte boundaries so every burst that reaches an intake maps to exactly one DRAM burst. The alignment mask is `DRAM_BURST_BYTES - 1`, where `DRAM_BURST_BYTES = BL * (DRAM_BEAT_WIDTH / 8)`. Splitting is transparent to the AXI master: a host burst that crosses a DRAM-burst boundary is decomposed into multiple aligned sub-bursts, each of which becomes one CAM entry / one DRAM command.

11.2 `pumice_wr_intake` / `pumice_rd_intake`

11.2.1 Purpose

Terminate the AXI channels (via the embedded `axi4_slave`), decode each transaction's address, and push a `{rank, bank, row, col, id}` insert request downstream into the corresponding CAM. The intakes are deliberately "dumb": there is no reordering or scheduling here — that lives in the scheduler layer reading the CAMs.

11.2.2 Write intake

- Accepts AW transactions and issues an `aw_push` insert into `pumice_wr_data_cam`.
- Streams W beats into the write-data FIFO tagged for the CAM SRAM fill mover.
- Returns the B response when the CAM signals commit-done for the transaction ID (`wr_done_valid` / `wr_done_id`), matching AXI4 posted-write semantics.

11.2.3 Read intake

- Accepts AR transactions and issues an `ar_push` insert into `pumice_rd_cmd_cam`.
- Before committing a miss to the read CAM, it probes the write CAM with a snarf query (`snarf_probe_*`). On a snarf hit the read is serviced directly from the write CAM's SRAM (read-your-write forwarding) rather than being scheduled to DRAM.
- Drains the read CAM (in AR order) onto the AXI R channel with the correct ID and `rlast`.

11.2.4 Per-ID Ordering

AXI4 requires reads from the same ID to return in order and writes from the same ID to commit in order. Because each intake inserts in AR/AW arrival order and the read CAM drains in insert

order, per-ID ordering is preserved. The CAMs allow row-hit scheduling to reorder DRAM commands across entries, but the AXI completion layer honors per-ID order.

11.2.5 Backpressure

- AW/AR `.ready` deassert when the corresponding CAM has no free entry (`ins_ready` low).
- W `.ready` deasserts when the write-data FIFO/SRAM fill path is full.
- R and B `.valid` follow standard AXI protocol; a stalled consumer stalls only the drain path and does not corrupt the controller core.

11.3 addr_mapper

11.3.1 Purpose

Translate a flat AXI address into DRAM coordinates `(rank, bank, row, col)`. RTL: `rtl/fub/addr_mapper.sv`. It is combinational and single-stage, instantiated inside each intake.

11.3.2 Interfaces

Inputs:

- `axi_addr_i` — `AXI_ADDR_WIDTH` bits
- `bank_lsb_i[4:0]` — the `ADDR_MAP.bank_lsb` CSR field
- `hash_en_i` — the `ADDR_MAP.hash_en` CSR field
- `hash_seed_i[7:0]` — the `ADDR_MAP.hash_seed` CSR field

Outputs:

- `rank_o` — `$clog2(NUM_RANKS)` bits
- `bank_o` — `$clog2(NUM_BANKS)` bits
- `row_o` — `ROW_WIDTH` bits
- `col_o` — `COL_WIDTH` bits

11.3.3 Mapping Function (single knob: bank_lsb)

There is **no scheme selector** anymore. The mapping is driven by one runtime knob, `ADDR_MAP.bank_lsb`, which places the bank field within the byte-offset-stripped word address (`word = axi_addr >> BYTE_OFFSET_WIDTH`). The column auto-splits around the bank; row and rank stack above the column region and their positions are **invariant**:

```
word address, low -> high:
[ col_lo(bank_lsb) | bank(BW) | col_hi(CW - bank_lsb) | row(RW) |
```

```
rank(KW) ]
col = { col_hi, col_lo }
row LSB is always at CW + BW (invariant — only the bank slides)
```

The RTL clamps `bank_lsb` to `[0, COL_WIDTH]` so the `col_hi` width stays non-negative and the row/rank slices land where the geometry expects.

11.3.4 Classic Schemes as `bank_lsb` Settings

The former named schemes are just settings of the one knob:

Effect	Setting	Notes
`ROW_MAJOR`	`bank_lsb == COL_WIDTH`	Bank above the whole column; default (`0x0A`)
max `BANK_INTERLEAVE`	`bank_lsb == log2(cols/burst)`	Minimal `col_lo` keeps a DRAM burst inside one bank
partial interleave	any value in between	Column splits around the bank

Software keeps `log2(cols/burst) <= bank_lsb <= COL_WIDTH` so a DRAM burst's column walk stays inside one bank.

11.3.5 Optional Bank XOR-Hash

When `hash_en` is set, the bank index is XOR-folded with row bits and the seed to defeat power-of-two-stride hot-banking (the former `XOR_HASH` scheme, now an orthogonal overlay on any `bank_lsb` placement):

```
bank[i] ^= row[i] ^ row[i + BW] ^ hash_seed[i]
```

The mid-slice index is clamped so it never runs past `ROW_WIDTH`.

11.3.6 Relationship to the DFI BFM

This module mirrors the `AddressMapping` class in the DFI BFM (`CocoTBFramework/components/dfi/`). The same `bank_lsb` / hash configuration drives both RTL elaboration-time verification and BFM checking, so the address decode is consistent between simulation and silicon.

12 Command Scheduler

The command-scheduling layer is `pumice_mem_cmd_scheduler` (`rtl/macro/pumice_mem_cmd_scheduler.sv`). It is a single controller-clock (`aclk`) layer that wires together the pick core and all the timing / bring-up support blocks, and emits one abstract DRAM command stream `{op, rank, bank, row, col, ap}` into a command FIFO for the DFI layer to pack onto phases.

The scheduler does **not** hold the transaction queue — pending requests live in the two CAMs inside `pumice_axi4_ifc`. The scheduler reads them through external lookup / oldest / commit / issue ports.

12.1 Composition

`pumice_mem_cmd_scheduler` instantiates:

- **`pumice_cmd_arbiter`** — the single pick core (open-page decision is inline).
- **`pumice_bank_timers`** (per-rank/bank `bank_timer` instances) — FSM-free per-bank JEDEC "safe" timers (see `03_bank_machines.md`).
- **`global_timers`** — cross-bank / bus turnaround (tFAW, tRRD, tWTR, tRTW, tCCD).
- **`refresh_ctrl`** — tREFI tracking + postpone accumulator (see `04_refresh.md`).
- **`init_sequencer`** — JEDEC MRS cold-boot init; gates traffic until done (see `05_init_power.md`).
- **`mode_register`** — CL / CWL / BL decode shadow, updated by init MRS writes.
- an output command FIFO (`gaxi_fifo_sync`).

12.2 `pumice_cmd_arbiter`

12.2.1 Purpose

Pick exactly one abstract DRAM command per cycle and push it into the scheduler-to-DFI command interface. It is PHY/nphases-agnostic and single-issue. JEDEC timing readiness is supplied by the per-bank (`pumice_bank_timers`) and global (`global_timers`) inputs; the arbiter never re-derives timing itself.

12.2.2 Inputs

- Per-bank readiness from `pumice_bank_timers`: `bank_act_ready`, `bank_rdwr_ready`, `bank_pre_ready`, `bank_row_active`, `bank_open_row`.

- Global readiness from `global_timers`: `tfaw_ok`, `trrd_ok` (per-rank), `twtr_ok`, `trtw_ok`, `tccd_ok`.
- Per-bank scheduler lookups into both CAMs (query each bank's open row), plus the CAMs' `oldest` ports.
- Refresh request / drain from `refresh_ctrl`; init command passthrough from `init_sequencer`.

12.2.3 Outputs

- Command push `{op, rank, bank, row, col, ap}` with `cmd_valid` / `cmd_ready`.
- Event strobes to the timers (`evt_act`, `evt_rd`, `evt_wr`, `evt_pre`, `evt_ap`, `evt_rank`, `evt_bank`, `evt_row`) — fired only on an accepted issue.
- CAM side effects: write-CAM commit, read-CAM issue, refresh grant — all gated on accepted issue.

`op` is drawn from `dram_op_e` (`OP_NOP`, `OP_ACT`, `OP_RD`, `OP_RDA`, `OP_WR`, `OP_WRA`, `OP_PRE`, `OP_PREA`, `OP_REF`, `OP_REFPB`, `OP_MRS`, …).

12.2.4 Priority Function

Evaluated combinationally each cycle (descending priority):

1. **Init in progress** (`!init_done`) — forward the `init_sequencer` command verbatim; block all normal traffic.
2. **Refresh** (`refresh_req` or drain active) — precharge active banks one per cycle, then — only under `w_ref_safe` (all rows closed in the registered view, nothing row-affecting in flight or guarded, prior REF's tRFC recovery elapsed) — issue `REF` and assert the refresh grant. Each fired REF loads the arbiter's tRFC down-counter (`TIMINGS_RFC_REFI.tRFC`); while non-zero, ACTs and further REFs are blocked.
3. **Column row-hit** — a `RD`/`WR` to an already-open row whose bank is column-ready (`bank_rdwr_ready`) and whose bus turnaround permits it (`tccd_ok` + `twtr_ok` for reads / `trtw_ok` for writes). **Reads have priority over writes**; within each, the **oldest** entry (max CAM relative age) wins the tie-break.
4. **Fallback** — no ready row-hit: `ACT` the oldest pending op's row on an idle bank (subject to `tfaw_ok` / `trrd_ok` and the ACT/PRE guard), or `PRE` a bank that is open on the wrong row.

The fallback target is read-priority: the read CAM's `oldest` port is consulted first, then the write CAM's.

12.2.5 ACT/PRE Re-Issue Guard

The bank timers register their readiness outputs (there is a two-cycle latency from an event to `act_ready` / `pre_ready` dropping). A stateless combinational arbiter would otherwise re-issue ACT/PRE to the same bank before the timers reflect it. The arbiter therefore keeps a two-cycle per-bank guard (`r_guard0` / `r_guard1`) that blocks a bank for two cycles after an accepted ACT, PRE **or column** to it — columns are included because a just-fired column has not yet dropped the bank's registered `pre_ready` (tRTP/tWR load), so an unguarded PRE (normal or refresh-drain) could precharge on stale readiness. Columns still self-limit against each other (both CAMs exclude a just-committed / just-issued slot from the next cycle's lookup), so the guard never gates column-vs-column. The shared-DQ-bus burst-occupancy constraint (a BL burst owns the bus for `BL/DFI_RATE` DFI cycles) is enforced downstream in the DFI command path, not here; the CDC decouples `aclk` command issue from the `dfi_clk` DQ timing.

12.2.6 Page Policy (Auto-Precharge)

The column auto-precharge bit is set directly from `page_policy_i`:

- **`OPEN`** — `ap = 0`; rows stay open. Column ops stream to an open row at tCCD rate.
- **`CLOSE`** — `ap = 1`; every column op auto-precharges (issues `RDA` / `WRA`).
- **`HAPPY_HYBRID`** — in v1 this is treated as `OPEN`. The `page_predictor` hook is a documented TODO; `rtl/fub/page_predictor.sv` exists but is not wired into the default build.

The open-page "keep the row open" decision therefore lives inline in the arbiter and the per-bank `bank_timer` (which keeps the row open on `RD`/`WR`), not in a separate predictor or lookahead unit.

12.2.7 Issue Rate

One command issue per controller clock cycle. Multi-phase / multi-cycle placement (including LPDDR2's 2-edge CA word) happens downstream in the DFI layer, so the arbiter always sees a single issue.

12.3 Mode Register Shadow

`mode_register` maintains the live CL / CWL / BL decode. Its shadow is written in lockstep by the `init_sequencer` MRS writes (`mr_seq_we` / `mr_seq_index` / `mr_seq_data`), so the controller's timing decode tracks exactly what was programmed into the DRAM during init. It exposes `cl_o` / `cwl_o` / `bl_o` to the DFI layer.

12.4 v1 Scope Notes

The arbiter picks from a single rank (rank 0) in v1; write-drain watermarking, multi-rank pick, and power-down coordination are documented TODOs. The `busy_o` output aggregates "init not done", "refresh pending", "command in the FIFO", and "any bank row active".

---

13 Bank Timers and Cross-Bank Timers

Per-bank JEDEC timing is tracked by an **FSM-free** `bank_timer` per (rank, bank); cross-bank / bus-turnaround timing is tracked by `global_timers`. There is no multi-state bank machine in this design.

13.1 `bank_timer`

13.1.1 Purpose

Track one DRAM bank's JEDEC "safe" timing. RTL: `rtl/fub/bank_timer.sv`. It is **not a state machine** — it is a set of preset/decrement countdown timers plus a trivial row-open register and a single auto-precharge bit. The arbiter drives one `set_*` strobe per issued command; each timer loads its config value on its trigger, free-runs decrementing (saturating at 0), and reports its constraint as "safe" when the count is 0.

Because the per-command `safe_*` outputs are a purely combinational AND of the relevant timers behind a **single register stage** (the counter), a just-issued command is reflected one cycle later with no multi-stage lag. This was the point of retiring the old 3-state FSM, whose double-registered state was the root cause of the refresh-vs-ACT and column-vs-PRE hazards.

13.1.2 Instantiation

`pumice_bank_timers` (`rtl/fub/pumice_bank_timers.sv`) stamps one `bank_timer` per `(rank, bank)` pair — for the default (`NUM_RANKS=1`, `NUM_BANKS=8`) that is 8 instances. It routes the arbiter's `evt_*` event strobes to the addressed bank and fans the per-bank readiness back out to the arbiter.

13.1.3 Timers

Each timer loads on its event, counts down, and saturates at 0:

Timer	JEDEC	Loaded on	Gates
r_rcd	tRCD	ACT	safe_rd/safe_wr
r_ras	tRAS	ACT	safe_pre
r_rc	tRC	ACT	safe_act
r_rp	tRP	explicit PRE or auto-PRE fire	safe_act
r_preblk	tRTP (RD) / tWR (WR)	RD / WR	safe_pre only

The timer config inputs (t_rcd_i, t_rp_i, t_ras_i, t_rc_i, t_wr_i, t_rtp_i) are controller-cycle, command-relative values supplied by the scheduler layer from CSRs.

13.1.4 Row State (minimal — not an FSM)

Two elements track row occupancy:

- r_row_valid + r_open_row — set on ACT (captures row_i), cleared on an explicit PRE or when an auto-precharge completes.
- r_ap_pending — a single bit set from the set_ap_i qualifier on a RD/WR. When both r_preblk and r_ras reach 0, the auto-precharge "fires" (w_ap_fire): the row closes internally and tRP loads — with no scheduler PRE and no extra states.

13.1.5 Combinational Safe Outputs

The readiness outputs are combinational off the timers (single register stage behind):

- safe_act_o = !r_row_valid && (r_rp == 0) && (r_rc == 0) — bank precharged, tRP since PRE, tRC since last ACT.
- safe_rd_o = r_row_valid && (r_rcd == 0) && !r_ap_pending — row open, tRCD met, not auto-precharging.
- safe_wr_o = safe_rd_o — same condition.
- safe_pre_o = r_row_valid && (r_ras == 0) && (r_preblk == 0) && !r_ap_pending — row open, tRAS and tRTP/tWR met, not auto-precharging.

Open-page behavior falls out naturally: RD/WR only load r_preblk and (optionally) the auto-PRE bit, so column commands stream to an open row. The tCCD / tWTR / tRTW turnaround constraints are enforced globally in global_timers and ANDed by the arbiter, not here.

13.1.6 Observability State

For debug only, `state_o` is derived combinationally from the timers into a `bank_state_e` value (`BANK_IDLE`, `BANK_ACTIVATING`, `BANK_ACTIVE`, `BANK_PRECHARGING`). No downstream logic depends on it — it is purely an observation output alongside the `obs_*_nz` timer-nonzero flags.

13.1.7 Refresh Interaction

There is no per-bank refresh handshake. The arbiter serializes refresh: on a refresh request it precharges active banks one per cycle (using `safe_pre` / `bank_row_active`), then issues `REF` once no bank has an open row AND `w_ref_safe` holds (no in-flight/guarded row op; prior tRFC recovery elapsed — see the scheduler chapter). REF recovery (tRFC) is an arbiter-side down-counter; the `bank_timer` never sees REF events. The `bank_timer` participates only through its standard row-open and `safe_pre` outputs.

13.2 global_timers

13.2.1 Purpose

Enforce the cross-bank and shared-bus turnaround constraints that a single bank cannot see locally. RTL: `rtl/fub/global_timers.sv`. Implemented once (not per bank) because these constraints are inherently global.

13.2.2 Tracked Constraints

Constraint	Description	Scope
`tFAW`	At most 4 ACTs in any rolling tFAW window	per-rank
`tRRD`	Minimum gap between any two ACT commands	per-rank
`tWTR`	Write-to-read data-bus turnaround	global
`tRTW`	Read-to-write data-bus turnaround	global
`tCCD`	Column-to-column spacing	global

13.2.3 Interface

The timers consume the arbiter's `evt_act` (with `evt_act_rank`), `evt_rd`, and `evt_wr` strobes and produce the readiness window signals the arbiter ANDs into its pick:

- `tfaw_window_ok_o` / `trrd_window_ok_o` — per-rank vectors, gate `ACT`.
- `twtr_global_ok_o` / `trtw_window_ok_o` / `tccd_window_ok_o` — gate column commands.

tFAW / tRRD are kept per-rank because they are device-local activate-stagger limits; the data-bus turnaround windows (tWTR / tRTW / tCCD) are global because the DQ bus is shared across ranks.

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14 Refresh Controller

`refresh_ctrl` (`rtl/fub/refresh_ctrl.sv`) owns tREFI timing and refresh request accounting. It does not issue commands itself — it raises a request to the command arbiter, which serializes the precharge-then-REF sequence and pulses a grant once REF reaches the DFI bus.

14.1 `refresh_ctrl`

14.1.1 Purpose

Track the interval between mandatory refreshes and tell the scheduler when refresh is owed. JEDEC permits up to 8 postponed refreshes; `refresh_ctrl` uses an 8-deep accumulator plus a drain-quota mechanism so a batch of owed refreshes can be granted back-to-back.

14.1.2 Interface

Signal	Direction	Purpose
`t_refi_i[15:0]`	input	Refresh interval in MC cycles (CSR-backed)
`refresh_burst_i[3:0]`	input	1..8 REFs to drain per request cycle (CSR-backed)
`refpb_mode_i`	input	0 = REFab, 1 = REFpb (LPDDR2) bank-rotor select
`enable_i`	input	Gates tREFI counting; driven by `init_done`
`refresh_req_o`	output	High while pending refreshes remain
`refresh_grant_i`	input	Pulsed by the arbiter when REF issues on the bus
`refresh_drain_active_o`	output	High during a back-to-back drain burst
`refresh_kind_o`	output	Registered REFab/REFpb selector
`refresh_bank_o`	output	Bank rotor value (valid in REFpb mode)

Signal	Direction	Purpose
pending_refreshes_o	output	Current accumulator value
obs_*	output	Internal state harvested for CSR readout

14.1.3 tREFI Counter and Pending Accumulator

r_refi_cnt counts down from t_refi_i. It only ticks while enable_i is high (i.e. after init completes); before that it is held reloaded at t_refi_i. On expiry it reloads and the pending accumulator r_pending increments by 1 (saturating at the JEDEC maximum of 8). Each accepted grant decrements r_pending. A simultaneous expiry and grant is a net-zero change. refresh_req_o stays high whenever r_pending > 0.

Saturating at 8 is JEDEC-conformant: the controller is expected to keep at most 8 postponed refreshes outstanding before it must catch up. If the system stays bandwidth-starved past 8 x tREFI, the counter saturates (a DRAM retention violation looming) rather than growing unbounded.

14.1.4 Drain Quota

r_burst_remaining implements back-to-back draining. When the previous burst is fully drained and pending work exists, it (re)loads min(refresh_burst_i, r_pending). Each grant decrements it. While r_burst_remaining > 0 and r_pending > 0, refresh_drain_active_o is asserted, which the arbiter reads as "keep granting REF back-to-back without yielding to reads/writes". Setting refresh_burst_i = 1 disables batching (one REF per tREFI).

14.1.5 REFpb Bank Rotor

When refpb_mode_i is set (LPDDR2 per-bank refresh), r_bank_rotor advances 0..NUM_BANKS-1 on each grant and is exposed on refresh_bank_o. In REFab mode it stays at 0. The current default build wires refpb_mode_i = 0 (REFab) in the scheduler.

14.2 Arbiter-Side Refresh Sequence

refresh_ctrl only raises refresh_req / refresh_drain. The precharge-then-REF sequence is performed by pumice_cmd_arbiter at refresh priority (second only to init):

1. While any bank on the target rank has an open row, precharge the active banks one per cycle (lowest ready bank first, honoring the ACT/PRE guard).
2. Once no bank has an open row **and w_ref_safe holds** — nothing row-affecting in flight or inside its 2-cycle guard window (the registered bank view alone is 2-3 cycles stale, which once let a REFab collide with a just-issued ACT), and the previous REF's tRFC recovery elapsed — issue OP_REF and assert refresh_grant_o to refresh_ctrl, decrementing the

accumulator / drain quota. On each fired REF the arbiter loads a **tRFC down-counter** from `TIMINGS_RFC_REFI.tRFC` (`t_rfc_i`); while non-zero, ACT picks and further REFs are blocked (mission-mode refresh recovery — init-time refreshes wait `INIT_TIMING1.t_rfc_wait` separately).

3. Repeat back-to-back while `refresh_drain_active` is high (each REF spaced by tRFC), until the accumulator is drained.

This "wait until all banks are precharged" behavior is implicit in the arbiter's readiness gating rather than an explicit bank-grant handshake.

14.3 Notes and Scope

- **Self-refresh** / **power-down** are handled by the separate `powerdown_ctrl` block (see `05_init_power.md`), not by `refresh_ctrl`.
- **PASR, DARP idle-bank-first selection, temperature-compensated tREFI, and periodic ZQCS** are not part of the current `refresh_ctrl` RTL. REFpb selection is a simple bank rotor; the richer selection policies remain future work.
- **REFab vs REFpb** for LPDDR2 is selectable via `refpb_mode_i` and the bank rotor, but the default build uses REFab.

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15 Init Sequencer and Power-Down

Cold-boot bring-up is handled by `init_sequencer`; low-power entry/exit by `powerdown_ctrl`.

15.1 `init_sequencer`

15.1.1 Purpose

Sequence post-reset DRAM bring-up: the DFI init handshake, JEDEC mode-register loads, precharge, and refresh. RTL: `rtl/fub/init_sequencer.sv`. It is a compact hard-coded FSM (not a microprogram ROM). Its commands are **issued to the DRAM**, not merely shadowed — this was the fix for the on-board bring-up failure where an earlier shadow-only walk left the read DLL unlocked.

15.1.2 Command Path

While `init_busy_o` is high, the scheduler stays idle and forwards the sequencer's single-cycle command pulses (`init_cmd_valid_o` / `init_cmd_op_o` / `init_cmd_bank_o` / `init_cmd_row_o`) straight to `dfi_cmd_formatter`. Each command state is occupied for exactly one cycle, then the FSM parks in `S_WAIT` for the JEDEC inter-command delay before advancing. Because `dfi_cmd_formatter` is always `cmd_ready`, a single-cycle pulse issues exactly one command with no grant handshake.

The `mode_register` shadow is updated in lockstep (`mr_seq_we_o` / `mr_seq_index_o` / `mr_seq_data_o`) so the live CL/CWL/BL decode tracks what was programmed.

15.1.3 CSR-Backed Waits

The inter-command delays are CSR-backed (INIT_TIMING registers), zero-extended into a 16-bit countdown, rather than hardcoded:

Input	JEDEC	Default
`t_init_wait_i`	CKE / tINIT	512
`t_dll_wait_i`	DLL lock tDLLK	256
`t_mrd_wait_i`	tMRD	8
`t_rp_wait_i`	tRP	8
`t_rfc_wait_i`	tRFC	8

15.1.4 DDR2 Sequence

Mirrors LiteDRAM's DDR2 PHY init (the reference proven on the Nexys A7 board):

1. Assert `dfi_init_start_o`; wait `dfi_init_complete_i` (the PHY runs its own DLL-lock / IO training), then wait tINIT.
2. Precharge All.
3. EMRS in JEDEC order EMRS(2) then EMRS(3) then EMRS(1) — all defaults 0.
4. MRS(0) + DLL reset (`MR0.VAL | 0x100`; reset default `0x533` = BL8/CL3/tWR3/DLL_RESET); wait tDLLK.
5. Precharge All.
6. Auto Refresh x2 (each followed by tRFC).
7. MRS(0) without DLL reset (`MR0.VAL`, reset default `0x433`) — clears the reset bit.
8. EMRS(1) + OCD default (`0x380`) then EMRS(1) + OCD exit (`0x000`).
9. `init_done_o = 1`.

The DDR2 MR values are CSR-backed (`MR0..MR3.VAL`; MR0 reset `0x433`, MR1 `0x0000`, MR2/MR3 `0x0000`) with the OCD default (`0x380`) ORed in by the sequencer; software may retune them before an `init_restart`.

15.1.5 LPDDR2 Sequence (fully functional)

Per JESD209-2F power-up and mode-register configuration:

1. Assert `dfi_init_start_o`; wait `dfi_init_complete_i`; tINIT settle.
2. MRW(MR63) = Reset (OP don't-care).
3. MRW(MR10) = `0xFF` ZQ Init Calibration.
4. MRW(MR1) = `0x23` (BL8 / nWR3), MRW(MR2) = `0x01` (RL3 / WL1), MRW(MR3) = `0x02` (DS 40 ohm).
5. `init_done_o = 1`.

Only MR1/MR2/MR3 update the CL/CWL/BL decode shadow; MR63/MR10 are issued to the DRAM but not shadowed (they exceed the 5-bit shadow index). The MR index (MA, up to MR63) and data (OP) are carried packed as `{MA[5:0], OP[7:0]}` in the ROW request field, so `dfi_cmd_formatter` can build the full LPDDR2 CA MRW word — a 3-bit bank port alone could not reach MR10/MR63.

15.1.6 Memtype Selection

`memtype_i` (from the PHY_TIMING CSR: 0 = DDR2, 1 = LPDDR2) chooses the sequence at runtime. After the shared DFI-init step, DDR2 branches to `S_PREA1` and LPDDR2 to `S_L_RESET`.

15.1.7 Status Outputs

Signal	Purpose
`dfi_init_start_o`	High after leaving reset
`init_busy_o`	High until the FSM reaches `S_DONE`
`init_done_o`	Asserted at `S_DONE`
`zqcl_req_o`	Tied off — DDR2 has no ZQCL

15.2 powerdown_ctrl

15.2.1 Purpose

Idle-detect into a low-power state. RTL: `rtl/fub/powerdown_ctrl.sv`. It is available but not in the default top build; it is documented here for completeness.

15.2.2 Modes

- **PDE (Precharge Power Down)** — CKE low, banks may stay active. Wakes on new activity within ~tXPDLL cycles.
- **SR (Self Refresh)** — CKE low plus an SRE command issued by the scheduler; the DRAM self-refreshes internally and the controller stops issuing REFs. Exit requires SRX plus tXSDLL (DDR2) / tXSR (LPDDR2) before new commands. SR entry requires all banks precharged first (the scheduler enforces the precondition).

15.2.3 Selection

- `enable_sref_i` high: on the idle threshold, request SR (`pdn_kind_o = 1`).
- `enable_sref_i` low, `enable_pde_i` high: request PDE (`pdn_kind_o = 0`).
- neither: never request.

15.2.4 Scope / TODO

Deep Power Down (LPDDR2), per-rank power-down, and the `dfi_dram_clk_disable` cooperation with `dfi_signal_pack` are documented TODOs. The block currently powers down all ranks together and trusts the scheduler not to grant before init completes.

---

16 Command Formatter and Signal Pack

Two modules translate the scheduler's abstract command into DFI wire signals: `dfi_cmd_formatter` (memtype-specific encoding and phase placement) and `dfi_signal_pack` (the final registered pipeline stage). `dfi_cmd_formatter` is instantiated inside `pumice_dfi_cmd_path` in the DFI layer.

16.1 `dfi_cmd_formatter`

16.1.1 Purpose

Translate the scheduler's chosen `dram_op_e` plus `(rank, bank, row, col)` into the multi-phase DFI v2.1 control bus. RTL: `rtl/fub/dfi_cmd_formatter.sv`. It is the only place where DDR2 and LPDDR2 differ at the wire layer, and its outputs are strict-flopped.

16.1.2 Input

The command arrives on `cmd_valid_i` / `cmd_ready_o` (always ready) with `cmd_op_i` (`dram_op_e`), `cmd_rank_i`, `cmd_bank_i`, `cmd_row_i`, `cmd_col_i`, `cmd_len_i`, and the runtime phase-placement knobs `rd_phase_i` / `wr_phase_i`.

16.1.3 Multi-Phase Output

For `DFI_RATE = N`, every control signal is packed as N phases side by side on the bus (`dfi_address_o`, `dfi_bank_o`, `dfi_cas_n_o`, `dfi_ras_n_o`, `dfi_we_n_o`, `dfi_cs_n_o`, `dfi_odt_o`). The decoded command is placed on its target phase and the other phases emit selected/deselected NOP. The target phase is:

- `rd_phase_i` for `OP_RD` / `OP_RDA`,
- `wr_phase_i` for `OP_WR` / `OP_WRA`,
- phase 0 for everything else (ACT/PRE/REF/MRS have no data-phase contract).

The R/W phase knobs match the PHY's rdphase/wrphase contract (e.g. a7ddrphy DDR2 CL3 nphases=2). Defaults of 0/0 preserve the legacy "everything on phase 0" behavior. `cs_n` is per-rank: `cs_n[r]=0` selects rank `r`; all-ones is a deselected NOP.

16.1.4 DDR2 Encoding

Combinational truth table per JESD79-2 Table 49:

op	cs_n	ras_n	cas_n	we_n	bank	addr
NOP	sel	1	1	1	-	-
ACT	sel	0	1	1	BA	row
RD	sel	1	0	1	BA	col (A10=0)
RDA	sel	1	0	1	BA	col + (1<<10)
WR	sel	1	0	0	BA	col
WRA	sel	1	0	0	BA	col + (1<<10)
PRE	sel	0	1	0	BA	A10=0
PREA	sel	0	1	0	-	(1<<10)
REF	sel	0	0	1	-	-
MRS	sel	0	0	0	MR	MR

op	cs_n	ras_n	cas_n	we_n	bank	addr	data

The auto-precharge / all-bank bit is A10. MRS data rides `cmd_row_i` (ROW_WIDTH), not `cmd_col_i` — MR0 = `0x533` needs bit 10, which a 10-bit column field would truncate.

16.1.5 LPDDR2 Encoding (bit-exact CA bus)

For LPDDR2 the command rides the multiplexed 10-bit CA bus over 2 edges, packed as a flat 20-bit word carried on `dfi_address` (low bits); `ras_n`/`cas_n`/`we_n` stay idle and `cs_n` asserts for the target rank. The two CA edges are already inside the word, so there is no per-DFI-phase command placement.

The CA word is built **bit-exact to JESD209-2F Table 60**, matching the DV BFM's `lpddr_ca` encoder. Layout: `w_lpddr2_ca[i] = CA{i}` rising edge (i = 0..9), `w_lpddr2_ca[10+i] = CA{i}` falling edge. Encoded commands include ACT, RD/RDA, WR/WRA, PRE, PREA, REF (all-bank), REFPB (per-bank), and MRW; NOP/Deselect drives CA0r..CA3r high. Column bit C0 is implied 0 and never transmitted; the auto-precharge flag lands on CA0f. The transcription reference is `rtl/LPDDR2_CA_ENCODING.md`.

For MRW, `MA0..MA5` map to `CA4r..CA9r`, `MA6/MA7` to `CA0f/CA1f`, and `OP0..OP7` to `CA2f..CA9f`. The init sequencer supplies the full MR index by packing `{MA[5:0], OP[7:0]}` into the ROW field, so MR10/MR63 are reachable.

> Note: the module's top-of-file header comment still carries a stale "LPDDR2 (TODO)" line from an earlier revision. The body implements the full bit-exact CA encoding above; LPDDR2 reads and writes are functional and pass the sim suite.

16.1.6 Idle Defaults

When not issuing, every phase is driven to its deselected idle: `cs_n=1, ras_n=1, cas_n=1, we_n=1, bank=0, address=0, odt=0` (DFI v2.1 Table 2 defaults). ODT is driven from this module — there is no standalone `odt_ctrl` block.

16.2 dfi_signal_pack

16.2.1 Purpose

Final pipeline-register stage on the DFI v2.1 bus. RTL: `rtl/fub/dfi_signal_pack.sv`. It latches every command / write-data / read-data-enable input and drives it out the next MC cycle, owning `dfi_dram_clk_disable` and reset-safe output values.

16.2.2 Behavior

- v1 is a pure one-cycle registered pipeline. Bus widths are `DFI_*_WIDTH * DFI_RATE`, so the multi-phase content from `dfi_cmd_formatter` passes through unchanged. This is where the internal `DW = DRAM_BEAT_WIDTH * DFI_RATE` word is presented across the `DFI_RATE` phases.
- `dram_clk_disable` is held at 0 in v1; per-phase staggering and power-down assertion are documented TODOs.

16.2.3 Multi-Cycle Commands

LPDDR2's 2-edge CA command is not split here. The 20-bit CA word is already packed onto `dfi_address` by `dfi_cmd_formatter`; the PHY splits it into two DRAM cycles per DFI v2.1.

16.2.4 Verification

The RTL output round-trips against the DFI BFM: the DDR2 truth table against the JESD79-2 reference, and the LPDDR2 CA bus against the shared `lpddr_ca` encoder/decoder (the same layout the RTL encodes).

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17 Write and Read Data Paths

The data path spans two clock domains. On the controller (`aclk`) side the two CAMs in `pumice_axi4_ifc` buffer burst data; on the DFI (`dfi_clk`) side the write serializer and read aligner drive/capture the DFI data bus. All four are **de-FSM'd streaming readers** — they are FIFO-fed / beat-counter datapaths with no active/slot state latch.

17.1 Controller-Side CAMs

17.1.1 `pumice_wr_data_cam`

RTL: `rtl/fub/pumice_wr_data_cam.sv`. A write-command CAM plus a write-data SRAM. Entries are keyed on `{bank, row, col}` with a free-running age; the burst payload lives in an SRAM slot. Three data movers stream over the SRAM:

- **fill** — captures W beats into the slot on insert.

- **commit-drain** — streams the slot to the DFI write path when the scheduler commits the entry.
- **snarf** — forwards a slot to a matching read (read-your-write), limited to unscheduled entries with a matching id and burst length.

Age is compared wrap-safe as a relative age (`age_ctr - entry_age`). The snarf lookup returns the **youngest** match (latest data), the scheduler lookup returns the **oldest** match (in-order commit per row), and the `oldest` port returns the oldest valid entry for the scheduler fallback. A fill-complete flag gates schedulability and snarf so a partially-filled burst is never committed or forwarded.

There is no standalone `wr2rd_forward` block — write-to-read forwarding is exactly the snarf mover in this CAM.

17.1.2 `pumice_rd_cmd_cam`

RTL: `rtl/fub/pumice_rd_cmd_cam.sv`. The read miss path — an outstanding-read tracker acting as a **reorder buffer**. Entries are keyed `{bank, row, col}` with a free-running age and N scheduler lookups so reads row-hit-schedule like writes. Data flows the opposite direction from the write CAM: it comes **in** from the DFI read return and drains **out** to `pumice_rd_intake`. DRAM read data returns in issue order and is buffered per entry; it drains in AR (insert) order, which is what enforces per-ID read ordering.

The issue-side oldest/lookups pick the oldest not-yet-issued entry; the drain side releases the oldest valid entry gated on data-ready. There is no snarf port (that is a write-CAM concept).

17.2 DFI-Side Data Movers

Both movers live in the DFI layer on `dfi_clk`. Their internal unit is the **DFI word** (`DRAM_BEAT_WIDTH * DFI_RATE` wide), which already carries all `DFI_RATE` phases; `dfi_signal_pack` splits it to the pins. One FIFO pop is one DFI cycle, so both are bubble-free.

17.2.1 `pumice_dfi_wr_serializer`

RTL: `rtl/fub/pumice_dfi_wr_serializer.sv`. On a `wr_fire` from the command path it waits `t_phy_wrlat_i` DFI cycles, then streams one DFI-word per cycle from the write-data FIFO onto `dfi_wrdata` (with `dfi_wrdata_en` and `dfi_wrdata_mask`) until the burst's `last` word. Because the write CAM pre-stages the burst into the FIFO when the command is scheduled, the burst is already waiting at `wr_fire` and the drive never stalls. The DFI byte mask is the inverted AXI strobe (`mask = ~strb`; AXI `wstrb=1` = write byte, DFI `mask=1` = mask).

17.2.2 `pumice_dfi_rd_aligner`

RTL: `rtl/fub/pumice_dfi_rd_aligner.sv`. The mirror of the write serializer. On a `rd_fire` it drives `dfi_rddata_en` for the read window starting `t_rddata_en_i` DFI cycles later, captures `dfi_rddata` whenever the PHY asserts `dfi_rddata_valid`, and pushes one DFI-word per valid

cycle into the read FIFO as `{last, resp, data}`. `BL_WORDS = BL/DFI_RATE` words per burst; `last` marks the final word so the read intake can split words back into AXI R beats. v1 supports one outstanding read window (single-issue, tRTW/tCCD spaced).

17.3 Response Generation

- **B response** — the write intake returns B when the write CAM signals commit-done for the transaction id, matching AXI4 posted-write semantics and decoupling B latency from DRAM-side delays.
- **R response** — the read intake drains the read CAM onto the R channel in AR order with the correct id and `rlast`. In v1 all reads return `OKAY` (the DFI `resp` field is carried through but DDR2/LPDDR2 have no CA parity to surface).

17.4 Runtime PHY Timing

The write and read timing offsets are runtime CSR knobs rather than hard-coded PHY latencies, so one build ports across PHYs: `t_phy_wrlat` (WR command to `dfi_wrdata_en`) and `t_rddata_en` (RD command to `dfi_rddata_en`), plus the `DFI_PHASE` CSR (`rd_phase` / `wr_phase`) covered in the DFI/CSR chapter.

---

18 DFI Layer and CSR

The bottom of the controller is the DFI layer (`pumice_dfi_layer`), which owns the single controller-to-PHY clock crossing, and the generated register block (`pumice_csr`), which carries all runtime configuration.

18.1 `pumice_dfi_layer`

18.1.1 Purpose

Present the controller-clock command / write-data / read-data streams on one side and the DFI v2.1 pin bus on the other, with exactly one clock-domain crossing between them. RTL: `rtl/macro/pumice_dfi_layer.sv`. The controller runs on `ctl_clk`; the PHY-facing datapath runs on `dfi_clk`.

18.1.2 The Single CDC

`pumice_dfi_cdc` (`rtl/fub/pumice_dfi_cdc.sv`) is the **only** clock-domain crossing in the whole controller. It is built from asynchronous `gaxi` FIFOs — three streams cross it:

- **cmd** (`ctl_clk -> dfi_clk`): the abstract command stream from the scheduler.
- **wrdata** (`ctl_clk -> dfi_clk`): DFI-word-granular `{last, strb, data}` drained from the write CAM.
- **rddata** (`dfi_clk -> ctl_clk`): `{last, resp, data}` captured from the PHY.
- plus the init handshake (`init_start` across, `init_complete` back).

The internal datapath unit is the DFI word (`dfi_wrdata` width, all `DFI_RATE` phases), so one FIFO word equals one DFI cycle and the crossing is bubble-free. Keeping the CDC in one place is what makes the rest of the controller a single-clock design.

18.1.3 DFI-Domain Datapath

On the `dfi_clk` side the layer instantiates three mechanical stages:

- **`pumice_dfi_cmd_path`** — pops the command FIFO and drives the multi-phase DFI command bus via `dfi_cmd_formatter` (see `06_encoder_output.md`). It emits `wr_fire` / `rd_fire` strobes to time the data movers and enforces the shared-DQ-bus burst occupancy (`COL_BURST_CYC = BL/DFI_RATE`).
- **`pumice_dfi_wr_serializer`** — drives `dfi_wrdata` at `t_phy_wrlat` (see `07_data_paths.md`).
- **`pumice_dfi_rd_aligner`** — captures `dfi_rddata` at `t_rddata_en` (see `07_data_paths.md`).

18.1.4 DFI Sub-Interfaces

The layer drives the three core DFI v2.1 sub-interfaces needed for DDR2 / LPDDR2 operation: the command bus, the write-data interface (`wrdata` / `wrdata_en` / `wrdata_mask`), and the read-data interface (`rddata_en` / `rddata` / `rddata_valid`), plus the DFI init handshake. Update / training / low-power / status sub-interfaces beyond `init_complete` are not driven; the controller manages power via CKE directly.

18.1.5 Runtime PHY Knobs

Rather than hard-coded PHY latencies, the layer takes runtime inputs so one build ports across PHYs: `rd_phase_i` / `wr_phase_i` (per-command DFI sub-phase placement), `t_phy_wrlat_i`, and `t_rddata_en_i`, plus `memtype_i`. These are driven from the CSR (see below).

18.2 `pumice_csr`

18.2.1 Purpose

Hold all runtime configuration and observation registers. The register block is **generated by PeakRDL** from `rtl/macro/pumice_csr.rdl` (via `bin/peakrdl_generate.py`), with a **passthrough cpuif** — it is not a hand-written APB3 slave.

18.2.2 Interface

The generated block exposes a simple request/response passthrough cpuif rather than a full bus protocol:

Signal	Direction	Purpose
`s_cpuif_req`	input	Request valid
`s_cpuif_req_is_wr`	input	Write (1) vs read (0)
`s_cpuif_addr`	input	Register address
`s_cpuif_wr_data[31:0]`	input	Write data
`s_cpuif_wr_biten[31:0]`	input	Per-bit write enable
`s_cpuif_rd_data[31:0]`	output	Read data
`s_cpuif_*_ack` / `*_err` / `*_stall_*`	output	Response / flow-control

Registers are `regwidth = 32` / `accesswidth = 32`. `pumice_top` connects the cpuif and fans the decoded `hwif_out` struct fields to the core; the core's status feeds back through `hwif_in`. Any external bus (APB, AXI-Lite) is adapted to this cpuif outside the block.

18.2.3 Access by Name

DV never uses hard-coded offsets. Configuration is programmed by field name through the generated `dv/tbclasses/pumice_regmap.py`, which stays in sync with the RDL. This is split-proof — if a register moves, the by-name access still resolves.

18.2.4 Address Mapping Register (ADDR_MAP @ 0x04C)

This register replaced the retired ADDR_MAP_TUNING (the old scheme_or / synth_mask_obs scheme selector is gone). It drives addr_mapper (see 01_axi_frontend.md):

Field	Bits	Default	Purpose
bank_lsb	[4:0]	0x0A	Bank field LSB in the word address (= COL_WIDTH -> ROW_MAJOR)
hash_en	[8]	0	Enable bank XOR-hash bank ^= fold(row) ^ hash_seed
hash_seed	[23:16]	0x00	XOR-hash seed

There is no scheme mux; bank_lsb alone selects row-major vs interleaved placement, and hash_en is an orthogonal overlay.

18.2.5 Key Configuration Registers

The RDL is authoritative; notable runtime-config registers include:

Register	Offset	Notable fields
DFI_PHASE	0x060	rd_phase[2:0], wr_phase[6:4] — DFI sub-phase placement (default 0/0)
PHY_TIMING	0x064	t_phy_wrlat[7:0] (reset 0; Nexys A7 tuple programs 1), t_rddata_en[15:8] (reset 6), memtype[16] (0=DDR2, 1=LPDDR2), refresh_burst[23:20] (1..8)
ADDR_MAP	0x04C	bank_lsb, hash_en, hash_seed (above)
INIT_TUNING	0x050	zq_retries[3:0], init_timeout_ms[15:8]
INIT_TIMING0	0x054 area	t_init_wait / t_dll_wait (JEDEC init waits, formerly hardcoded)

memtype selects DDR2 vs LPDDR2 at runtime and is consumed by init_sequencer, dfi_cmd_formatter, and mode_register. Confirm every register and field against the RDL and the generated regmap.

18.2.6 Narrow-Device (x16) Width Granularities

The controller keeps two width concepts distinct: the **DRAM beat** (one DFI phase's data, `DRAM_BEAT_WIDTH`) and the **physical device word** (the DQ width, `DRAM_DEVICE_WIDTH`, e.g. 16 for a x16 device). When a beat is wider than the device, burst length (beats per command) and the column-address stride are device-word quantities. The RTL scales the MR0 burst length by `DRAM_BEAT_WIDTH / DRAM_DEVICE_WIDTH` and sizes the column granularity to the device word; missing either produced the on-silicon DDR2 read failure on the Nexys A7 x16 bring-up (over-read plus column-overlap scramble). PHY read-path integration (data-vs-`rddata_valid` skew and per-command read/write phase) is handled by the runtime `DFI_PHASE` / `PHY_TIMING` knobs above rather than hard-coded PHY latencies, so one controller build ports across PHYs.

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19 AXI4 Slave Interface

The controller exposes a standard AXI4 slave (`pumice_top` / `pumice_core`) for SoC connectivity. All slave ports use the `s_axi_*` prefix with no `i_`/`o_` decoration (e.g. `s_axi_awid`, `s_axi_awaddr`, `s_axi_wdata`, `s_axi_rdata`).

The AXI side runs on `aclk` / `aresetn`.

19.1 Data Width Is Fixed at DW

The core's AXI data width is **not** a free parameter. It is fixed at

```
DW = DRAM_BEAT_WIDTH * DFI_RATE
```

i.e. one AXI beat carries exactly one DFI word (= `DFI_RATE` DRAM beats). There is no independent `AXI_DATA_WIDTH` parameter on `pumice_top`; `s_axi_wdata` / `s_axi_rdata` / `s_axi_wstrb` are always `DW` / `DW` / `DW/8` bits.

To attach a host AXI bus of a *different* width, use the optional wrapper `pumice_top_geared` (see Host-Width Gearing below).

19.1.1 Hard width rule (compile-enforced)

`pumice_top_geared`'s `HOST_AXI_DATA_WIDTH` and the DFI word `DW` **MUST** relate by an **exact power-of-two ratio** — `AXI:DFI = G:1` or `1:G`, `G ∈ {1,2,4,8,…}`. An `initial assert … $fatal` in `pumice_top_geared` **fails elaboration** / **Vivado synthesis** on any other pairing, so a bad width choice is a **compile error**, never a silent broken build. Rationale (identical to

LiteDRAM, whose AXI frontend is 1:1 with its native port and routes all width change through a power-of-two stride converter): the DFI word is the atomic memory-side transfer, so an AXI beat must be a whole power-of-two number of DFI words. `DW` and `HOST_AXI_DATA_WIDTH` are the **only** compile-time width parameters — gear (active phases) and burst length are runtime CSRs (build-for-max).

19.2 Channels

Standard AXI4 — all five channels (AW, W, B, AR, R) are mandatory. No exclusive access support; `s_axi_awlock` / `s_axi_arlock` are present but ignored.

19.2.1 AW (Write Address) — SoC master → controller slave

Signal	Width	Notes
`s_axi_awid`	`AXI_ID_WIDTH`	Transaction ID
`s_axi_awaddr`	`AXI_ADDR_WIDTH`	Flat byte address
`s_axi_awlen`	8	Burst length minus 1
`s_axi_awsize`	3	Beat size; up to bus width
`s_axi_awburst`	2	`INCR` mandatory; `FIXED`/`WRAP` optional
`s_axi_awlock`	1	Present; ignored (no exclusive access)
`s_axi_awcache`	4	Present; ignored
`s_axi_awprot`	3	Present; ignored
`s_axi_awqos`	4	Present; ignored
`s_axi_awregion`	4	Present; ignored
`s_axi_awuser`	1	Present; ignored
`s_axi_awvalid`	1	Master asserts
`s_axi_awready`	1	Controller asserts when the WR intake / CAM can accept

19.2.2 W (Write Data)

Signal	Width	Notes
`s_axi_wdata`	`DW`	Beat data (`DW = DRAM_BEAT_WIDTH*DFI_RATE`)
`s_axi_wstrb`	`DW/8`	Byte enables; inverted to `dfi_wrdata_mask`
`s_axi_wlast`	1	Final beat of burst
`s_axi_wuser`	1	Present; ignored
`s_axi_wvalid`	1	Master asserts

Signal	Width	Notes
s_axi_wready	1	Controller asserts while the WR data path can accept

Note: AXI4 W channel has no wid; beats are presumed in the order their AW transactions were issued. The WR intake associates write data with its AW by arrival order.

19.2.3 B (Write Response)

Signal	Width	Notes
s_axi_bid	AXI_ID_WIDTH	Matches awid
s_axi_bresp	2	OKAY or SLVERR
s_axi_buser	1	Present; ignored
s_axi_bvalid	1	Controller asserts on response ready
s_axi_bready	1	Master asserts

19.2.4 AR (Read Address)

Signal	Width	Notes
s_axi_arid	AXI_ID_WIDTH	Transaction ID
s_axi_araddr	AXI_ADDR_WIDTH	Flat byte address
s_axi_arlen	8	Burst length minus 1
s_axi_arsize	3	Beat size
s_axi_arburst	2	INCR mandatory; others optional
s_axi_arlock	1	Present; ignored
s_axi_arcache	4	Present; ignored
s_axi_arprot	3	Present; ignored
s_axi_arqos	4	Present; ignored
s_axi_arregion	4	Present; ignored
s_axi_aruser	1	Present; ignored
s_axi_arvalid	1	Master asserts
s_axi_arready	1	Controller asserts when the RD intake / CAM can accept

19.2.5 R (Read Data)

Signal	Width	Notes
s_axi_rdata	DW	Beat data
s_axi_rid	AXI_ID_WIDTH	Matches arid
s_axi_rresp	2	OKAY (v1 always)
s_axi_rlast	1	Final beat of burst
s_axi_ruser	1	Present; ignored
s_axi_rvalid	1	Controller asserts when beat ready
s_axi_rready	1	Master asserts

19.3 Ordering Rules

- **Within an ID:** AXI4 ordering preserved. Reads from `arid = X` return in AR order; writes to `awid = X` commit in AW order.
- **Across IDs:** Out-of-order completion permitted by default (`AXI_OOO_ACROSS_IDS = true`). Disabling forces in-order across IDs at the cost of bandwidth.

19.4 Burst Type Support

Burst	Mandatory	Notes
FIXED	optional	Enable with SUPPORT_FIXED_BURST; rarely useful for DRAM
INCR	mandatory	Standard incrementing burst
WRAP	optional	Enable with SUPPORT_WRAP_BURST

19.5 Burst Length and DRAM-Burst Geometry

AXI4 allows up to 256 beats. The frontend `pumice_axi4_ifc` splits every host burst at DRAM-burst-byte boundaries (`axi_master_wr_splitter` / `axi_master_rd_splitter`) so each split request maps to exactly one DRAM burst.

At the DW side the required geometry is:

```
(awlen + 1) * DFI_RATE == BL
```

i.e. one AXI burst at DW equals one DRAM burst of BL beats. (The older "AXI_DATA_WIDTH == DRAM_BEAT_WIDTH" coupling no longer applies — the AXI beat is a full DFI word of DFI_RATE DRAM beats.)

19.6 Narrow Transfers

AXI4 supports awsize < bus width. The narrow case is handled via wstrb — the unused byte lanes are masked. dfi_wrdata_mask = ~wstrb, so the DRAM commits only the unmasked bytes.

19.7 Write Response Timing

The B response is **commit-driven**: it is returned once the write is accepted and committed by the controller (pumice_wr_intake raises B from the downstream commit-done handshake). A SLVERR B is self-generated by the intake for a malformed request; otherwise OKAY.

19.8 Frontend and Backpressure

The AXI frontend is pumice_axi4_ifc: dumb 1:1 write and read intakes (pumice_wr_intake, pumice_rd_intake) feeding a write-data CAM (pumice_wr_data_cam, the write buffer / snarf source) and a read reorder CAM (pumice_rd_cmd_cam), with in-order commit. There is no monolithic txn_queue.

- AW / AR .ready deassert when the corresponding intake or CAM cannot accept a new request (insertion ready low).
- W .ready deasserts when the write-data path (CAM buffer) is full.
- R / B .valid follow standard protocol; consumers may stall indefinitely without controller misbehavior.

19.9 Host-Width Gearing

To use a host AXI data width different from DW, instantiate the optional wrapper pumice_top_geared (parameter HOST_AXI_DATA_WIDTH, verified at {64, 128, 256}). It inserts the repo's formally-verified axi4_dwidth_converter_wr / axi4_dwidth_converter_rd between the host-width slave and the DW-width core. When HOST_AXI_DATA_WIDTH == DW, a generate branch bypasses the converters entirely, so the build is bit-identical to pumice_top (no converter, no added latency). See docs/AXI_DRAM_GEARING_SCOPE.md.

19.10 AXI4 Reset

aresetn is the AXI reset (active-low). When asserted, all AXI-side state is cleared. aclk / aresetn clock and reset the controller/AXI domain; the DFI side uses the independent dfi_clk / dfi_rstn (see the DFI chapter).

20 DFI v2.1 Master Interface

The controller drives DFI v2.1 master signals to the PHY. DFI v2.1 is the first DFI revision that supports LPDDR2; this controller targets that revision.

20.1 Gearing and Per-Phase Packing

The gearing / phase count is set by the parameter `DFI_RATE` (default 2), **not** `N_PHASES`. Rather than the DFI `_pN` suffix convention, this controller packs the `DFI_RATE` sub-phases **flat** into each bus. All DFI pins carry the `_o`/`_i` direction suffix (e.g. `dfi_address_o`, `dfi_rddata_i`).

Widths (top-level `pumice_top`, derived from geometry):

- `dfi_address_o` — `DFI_ADDR_BUS_W = ROW_WIDTH * DFI_RATE`
- `dfi_bank_o` — `$clog2(NUM_BANKS) * DFI_RATE`
- `dfi_cas_n_o` / `dfi_ras_n_o` / `dfi_we_n_o` — `1 * DFI_RATE`
- `dfi_cs_n_o` / `dfi_odt_o` — `NUM_RANKS * DFI_RATE`
- `dfi_wrdata_o` / `dfi_rddata_i` — `DFI_DATA_WIDTH = DRAM_BEAT_WIDTH * DFI_RATE`
- `dfi_wrdata_mask_o` — `DFI_DATA_WIDTH/8`
- `dfi_wrdata_en_o` / `dfi_rddata_en_o` — `DFI_RATE`
- `dfi_rddata_valid_i` — `DFI_RATE`

Per-phase slice `p` of a bus occupies `[p*W_phase +: W_phase]`.

Command placement (see `dfi_cmd_formatter.sv`): the decoded DDR2 command is placed on a target sub-phase — READ on `rd_phase`, WRITE on `wr_phase` (both CSR-driven via the `DFI_PHASE` register), and all other commands (ACT / PRE / REF / MRS) on phase 0. Non-driven phases emit NOP. This aligns R/W commands with the PHY's `rdphase` / `wrphase` contract.

20.2 Command Interface (MC → PHY)

Signal	Width	Notes
`dfi_address_o`	`ROW_WIDTH*DFI_RATE`	DDR2: row / column per phase; LPDDR2: 20-bit packed CA word on the low bits

Signal	Width	Notes
dfi_bank_o	clog2(NUM_BANKS)*DFI_RATE	DDR2: bank; LPDDR2: held idle
dfi_cas_n_o	DFI_RATE	DDR2: CAS encoding; LPDDR2: held idle
dfi_ras_n_o	DFI_RATE	DDR2: RAS encoding; LPDDR2: held idle
dfi_we_n_o	DFI_RATE	DDR2: WE encoding; LPDDR2: held idle
dfi_cs_n_o	NUM_RANKS*DFI_RATE	Per-rank chip select; cs_n[r]=0 selects rank r
dfi_odt_o	NUM_RANKS*DFI_RATE	On-die termination

There is no dfi_cke or dfi_reset_n port on the controller top; CKE / reset sequencing is not part of this pin bus.

20.2.1 DDR2 Command Encoding (JESD79-2)

The DDR2 truth table implemented by dfi_cmd_formatter.sv:

op	cs_n	ras_n	cas_n	we_n	bank	addr[10]	addr[..0]
NOP	0	1	1	1	X	X	X
ACT	0	0	1	1	BA	row	row
RD	0	1	0	1	BA	0	col
RDA	0	1	0	1	BA	1 (auto-PRE)	col
WR	0	1	0	0	BA	0	col
WRA	0	1	0	0	BA	1 (auto-PRE)	col
PRE	0	0	1	0	BA	0 (single bank)	X
PREA	0	0	1	0	X	1 (all banks)	X
REF	0	0	0	1	X	X	X
MRS	0	0	0	0	MR idx	MR data	MR data

cs_n is per-rank: cs_n[r]=0 selects rank r; all-ones deselects (NOP). A10 (addr[10]) is the auto-precharge bit for RD/WR and the all-bank bit for PRE. MRS carries its mode-register data on the row field (cmd_row_i, ROW_WIDTH), not the column field, so the full DDR2 MR bit range is preserved.

20.2.2 LPDDR2 Command Encoding (JESD209-2F)

LPDDR2 is fully functional (reads and writes). LPDDR2 has no RAS/CAS/WE strobes: the command and a scrambled address are multiplexed onto the 10-bit CA bus over two clock edges. `dfi_cmd_formatter.sv` builds the bit-exact JESD209-2F Table 60 encoding as a flat **20-bit** word (10 bits rising + 10 bits falling) placed on the low 20 bits of `dfi_address_o`; `dfi_ras_n_o` / `dfi_cas_n_o` / `dfi_we_n_o` are held idle and `dfi_cs_n_o` is asserted for the target rank. The full pin layout is the locked spec in `rtl/LPDDR2_CA_ENCODING.md`, which both the RTL and the DV BFM encode against.

20.3 Write Data Interface (MC → PHY)

Signal	Width	Notes
`dfi_wrdata_o`	`DFI_DATA_WIDTH`	Write data (one DFI word = all phases)
`dfi_wrdata_en_o`	`DFI_RATE`	High during valid write-data phases
`dfi_wrdata_mask_o`	`DFI_DATA_WIDTH/8`	`mask = ~wstrb` (1 = mask byte)

Write data is driven by `pumice_dfi_wr_serializer`: on a WR command it waits `t_phy_wrlat` DFI cycles, then streams one DFI word per cycle from the write-data FIFO until the burst's last word (bubble-free, back-to-back bursts).

20.4 Read Data Interface (PHY ↔️ MC)

Signal	Width	Direction	Notes
`dfi_rddata_i`	`DFI_DATA_WIDTH`	PHY → MC	Read data (one DFI word)
`dfi_rddata_valid_i`	`DFI_RATE`	PHY → MC	High when a read word is valid
`dfi_rddata_en_o`	`DFI_RATE`	MC → PHY	Read capture window

Read alignment is done by `pumice_dfi_rd_aligner`: on a RD command it asserts `dfi_rddata_en_o` for the burst's word count starting `t_rddata_en` DFI cycles later, and captures `dfi_rddata_i` on each `dfi_rddata_valid_i`. There is no `dfi_rddata_dnv` port.

20.5 Status / Init Interface

Signal	Width	Direction	Notes
`dfi_init_start_o`	1	MC →	Controller requests PHY init start

Signal	Width	Direction	Notes
		PHY	
`dfi_init_complete_i`	1	PHY → MC	High when the PHY has finished init

These are the only status handshake pins on the controller top. Update and training sub-interfaces are not present on this pin bus.

20.6 Timing Parameters Honored

The DFI datapath (`pumice_dfi_layer`) honors these CSR-driven knobs:

- `t_phy_wrlat` — delay from the WR command to `dfi_wrdata_en` (write serializer)
- `t_rddata_en` — delay from the RD command to the `dfi_rddata_en` capture window (read aligner)
- `rd_phase` / `wr_phase` — the DFI sub-phase carrying the READ / WRITE command (`DFI_PHASE` register)

20.7 Clock Domains and CDC

The controller/AXI side runs on `aclk`; the PHY/DFI side runs on the independent input clock `dfi_clk` (with `dfi_rstn`). The two are **not** related by a fixed frequency ratio and there is no combinational phase replication.

The single clock-domain crossing lives inside `pumice_dfi_layer` (in `pumice_dfi_cdc`) and is built from **asynchronous gaxi FIFOs** — one each for the command, write-data, and read-data streams (plus the init handshake). The internal datapath unit is one DFI word (= one DFI cycle), so one FIFO word maps to one DFI cycle and the crossing is bubble-free.

---

21 CSR Register Interface (PeakRDL cpuif)

Configuration and observation registers are exposed through a **PeakRDL-generated register block** (`pumice_csr`), not a hand-written APB slave. The register map is authored in

`rtl/macro/pumice_csr.rdl` and generated with the passthrough cpuif (`bin/peakrdl_generate.py`); `pumice_top` instantiates the generated `pumice_csr` and drives the core by name from `hwif_out.*` (see §6.3 for the full map). The DV mirror used for by-name access is `dv/tbclasses/pumice_regmap.py`.

The SoC-facing bus is therefore the PeakRDL **passthrough cpuif**, a simple request/ack register interface. An SoC that natively speaks APB / AXI-Lite places a thin protocol adapter in front of this cpuif; the controller itself does not embed one in this generation.

21.1 cpuif Signals

`CSR_ADDR_W = 12` (4 KB, 32-bit registers). The block runs on `aclk` / `~aresetn` (the controller clock/reset — see below).

Signal	Width	Direction	Notes
`s_cpuif_req`	1	input	Request valid
`s_cpuif_req_is_wr`	1	input	1 = write, 0 = read
`s_cpuif_addr`	12	input	Register byte address (4 KB space)
`s_cpuif_wr_data`	32	input	Write data
`s_cpuif_wr_biten`	32	input	Per-bit write enable
`s_cpuif_req_stall_wr`	1	output	Back-pressure a write request
`s_cpuif_req_stall_rd`	1	output	Back-pressure a read request
`s_cpuif_rd_ack`	1	output	Read data valid
`s_cpuif_rd_err`	1	output	Read error (e.g. unmapped address)
`s_cpuif_rd_data`	32	output	Read data
`s_cpuif_wr_ack`	1	output	Write accepted
`s_cpuif_wr_err`	1	output	Write error

21.2 Access Semantics

Standard PeakRDL passthrough behavior:

- Single-cycle accepts for the typical case; the `*_stall_*` outputs allow the block to hold a request when needed.
- `s_cpuif_rd_err` / `s_cpuif_wr_err` are raised for unmapped addresses; the error terminates the access and does not disturb controller state.

- Read-only fields ignore writes; write-only / self-modifying fields (e.g. `CTRL.init_start`, `CTRL.soft_reset`) pulse `swmod`.

21.3 Clock Domain

The register block is clocked by `aclk` (the controller / host-AXI clock) and reset by `~aresetn`. It is **not** on a separate management clock in this generation — there is no independent `apb_pclk`. Config fields (`hwif_out.*`, `hw = r`) are read combinationally by `pumice_top` and fanned out to the core; the single clock-domain crossing to the PHY side (`dfi_clk`) lives inside `pumice_dfi_layer` (async gaxi FIFOs), not in the register block. Observation fields (`hwif_in.*`, `hw = w`) are written from the core clock domain.

21.4 Register Map

The full address map is documented in §6.3, reconciled against `rtl/macro/pumice_csr.rdl` and `dv/tbclasses/pumice_regmap.py`.

21.5 Configuration Sequencing

Recommended bring-up order (see §6.1 for reset details):

1. Hold `aresetn` asserted.
2. Program timing parameters (`TIMINGS_*`), MR overrides, and `PHY_TIMING` (including `memtype` and `t_phy_wrlat` / `t_rddata_en`).
3. Program `ADDR_MAP` (`bank_lsb` / `hash_en` / `hash_seed`) and `DFI_PHASE`.
4. Configure PASR mask (LPDDR2 only).
5. Release `aresetn`.
6. Wait for `STATUS.init_done`.
7. Begin AXI traffic.

Runtime tuning writes (scheduler / refresh / page policy) take effect at the next configuration quiet point; the SoC owns the drain — see §5.4.

22 Build-Time vs. Runtime Configuration

This controller distinguishes between **build-time parameters** that set the silicon footprint and **runtime CSR registers** that select behavior within that footprint. The two layers serve different purposes and should not be confused. This chapter applies the principle to every configurable knob in the design.

22.1 Principle

Layer	What it controls	Frozen at	Cost of changing
Build-time parameter	Synthesized silicon footprint: signal widths, table depths, queue sizes, instantiation counts, encoder selection, MAX bounds of runtime values	Elaboration time	Full RTL rebuild + place-and-route + tape-out (silicon) or bitstream rebuild (FPGA)
Runtime CSR register	Behavior selection *within* the synthesized footprint: policy choices, threshold values, on/off toggles, active counts up to the build-time MAX	Programmable any time the configuration quiet point is reached (typically during init or under SoC orchestration)	A single APB write

The architectural rule is straightforward: **if a knob is changing silicon area, it must be build-time. If a knob is only changing behavior of already-synthesized hardware, it must be runtime.**

22.2 Why This Distinction Matters

A controller designed without this distinction tends to collapse one of two ways:

- **Too much build-time** — every algorithmic choice is a parameter that requires a new bitstream / silicon revision to evaluate. Characterization

becomes impossibly slow; post-silicon debug becomes impossible. Most academic prototype controllers fall here.

- **Too much runtime** — every option is always synthesized, including ones that will rarely be used. Area balloons; timing closure suffers; the design becomes harder to verify because every code path is live. This is the failure mode of "ultra-configurable" commercial IP.

This controller targets the middle: **build-time picks what's physically present; runtime picks what's active.**

22.3 The Hybrid Pattern: "Build-Time MAX + Runtime Active"

Many of this controller's knobs follow a hybrid pattern where the build-time parameter sets a synthesized MAX, and a paired runtime CSR field selects the actually-used value within that MAX.

Concrete example for `LOOKAHEAD_DEPTH`:

```
Build-time:    LOOKAHEAD_DEPTH_MAX = 4
               → synthesizes 4 row-address comparators per bank
machine

Runtime CSR:   sched_tuning.lookahead_active [3:0]
               → scheduler peeks min(LOOKAHEAD_DEPTH_MAX,
lookahead_active)
                 same-bank requests per cycle

For debug:     Write lookahead_active = 0 to force pure HAPPY/page-
policy mode
               Write lookahead_active = 4 to use the full synthesized
window
               No rebuild required.
```

The same pattern applies to `TXN_QUEUE_DEPTH`, `AGE_MAX`, `REFRESH_DEFER_MAX`, `INIT_ZQ_RETRIES`, `ZQCS_FREQ_HZ`, and others — area is paid once at synthesis for the MAX; runtime selects how much of the synthesized hardware is actively gated.

22.4 The "Build-Time + Runtime Mux" Pattern

For algorithmic choices where multiple alternatives have similar area cost, the build-time parameter selects **which** code paths are synthesized, and a runtime CSR field picks **which** of the synthesized paths is active.

Concrete example for `PAGE_POLICY`:

```
Build-time:    PAGE_POLICY = HAPPY_HYBRID
               → synthesizes OPEN, CLOSE, and HAPPY decision logic
                 (plus the HAPPY predictor table)

Runtime CSR:   refresh_tuning.page_policy_or [1:0]
               00 = use build-time default (HAPPY_HYBRID here)
               01 = force OPEN
               10 = force CLOSE
               11 = force HAPPY_HYBRID

For debug:     Sweep policies on real workloads with no rebuilds.
```

The build-time choice `PAGE_POLICY = OPEN` would have omitted the HAPPY predictor table from the synthesized design entirely, saving area but losing the ability to switch to HAPPY at runtime. The build-time choice `PAGE_POLICY = HAPPY_HYBRID` synthesizes everything, paying the area cost for the predictor table, in exchange for the ability to switch among any of the three at runtime.

The same pattern applies to `REFPB_POLICY`, `SCHEDULER_MODE`, and HAPPY enable/disable.

Note that address mapping is **not** an example of this pattern. Earlier revisions exposed an `ADDR_MAP_SCHEME` build-time set plus an `ADDR_MAP_TUNING.scheme_or` runtime mux; both have been retired. Address mapping is now a single runtime placement knob — see "The 'Runtime Placement' Pattern" below.

22.5 The "Runtime Placement" Pattern

Some knobs neither gate synthesized alternatives nor cap a synthesized MAX — they simply reposition a field within already-present combinational logic. There is no mux and no area choice; the same decode hardware produces different behavior depending on a runtime value.

Address mapping is the canonical case. `addr_mapper.sv` decodes a flat AXI address into `{rank, bank, row, col}` from **one** runtime CSR field, `ADDR_MAP.bank_lsb`, which selects where the bank field sits in the byte-offset-stripped word address. The column fills below (`col_lo`) and above (`col_hi`) the bank; row/rank stack above at invariant positions. The classic "schemes" are just settings of `bank_lsb`:

```
Runtime CSR:   ADDR_MAP.bank_lsb [4:0]   (reset = 0x0A = COL_WIDTH =
ROW_MAJOR)
               = COL_WIDTH               -> bank above whole column =
ROW_MAJOR
               = log2(BL/DFI_RATE)       -> minimal col_lo          =
BANK_INTERLEAVE
               in between                -> partial interleave

Runtime CSR:   ADDR_MAP.hash_en [8], ADDR_MAP.hash_seed [23:16]
```

```
folds row bits + seed into the bank index (= XOR_HASH),
defeating power-of-two-stride hot-banking.
```

There is no build-time parameter and no scheme selector: `ROW_MAJOR`, `BANK_INTERLEAVE`, and `XOR_HASH` are all reachable by writing `bank_lsb` (and optionally `hash_en`/`hash_seed`) at runtime. The address-decode logic is combinational and always present.

`MEMTYPE` is likewise a runtime knob in this generation, not a build-time string. The core decodes `memtype_e` from the CSR field `PHY_TIMING.memtype` (`0 = DDR2`, `1 = LPDDR2`) — see `pumice_top.sv` `w_memtype`. There is no `MEMTYPE` build parameter that gates an encoder or step table; the DDR2 vs LPDDR2 behavior is selected at runtime within the synthesized datapath.

22.6 The "Runtime Only" Pattern

A small number of knobs have zero area cost and are runtime-only: simple counter compare values, timeout thresholds, retry counts. Examples include `ZQCS_FREQ_HZ`, `INIT_ZQ_RETRIES`, and the various observation-counter clear-on-read controls.

These have no build-time presence at all.

22.7 Configuration Quiet Points

Runtime CSR writes that change scheduler or refresh-manager behavior take effect at the next **configuration quiet point** — defined as: no DRAM commands in flight, no refresh sequence in progress, and the relevant FSM in its idle state. The CSR slave does not enforce quiet-point writes; the SoC firmware is responsible for sequencing. Writing during normal operation is well-defined but the change does not apply until the next quiet point. There is **no** `config_apply` strobe in this generation — the SoC owns the drain and must quiesce host AXI traffic (and, if needed, wait out any in-flight refresh) before the change is guaranteed to take effect.

Writes to register-file CSRs that have no FSM impact (timing parameter overrides, MR override values, observation counters) take effect immediately and have no quiet-point requirement.

22.8 Summary

The following two chapters document this controller's complete configuration interface:

- **§5.2 Top-Level Parameter Table** — every build-time parameter, with its type, range, governing section, and (where applicable) the paired runtime CSR field.
- **§5.3 Characterization Knobs** — the subset of build-time + runtime knobs intentionally exposed for performance sweeping during silicon / FPGA characterization.

The full runtime CSR register map is in §6.3.

23 Top-Level Parameter Table

All build-time parameters for the controller, with type, default, valid range, the section that governs their use, and a one-line purpose summary.

23.1 Parameter Table

The parameters below are the actual elaboration parameters of `pumice_top` / `pumice_core` (with the host-width parameter added by `pumice_top_geared`). Several knobs that earlier revisions listed as build-time — `MEMTYPE`, `WRPHASE`, `RDPHASE`, and the address-map scheme set — are runtime CSR fields in this generation and are called out under “Runtime CSR knobs (not build parameters)” below.

Parameter	Type	Default	Range / valid set	Section	Purpose
`AXI_ID_WIDTH`	int	8	1-16	§3.1	AXI ID tag width (`IW`).
`AXI_ADDR_WIDTH`	int	32	24-40	§3.1	AXI flat address width (`AW`).
`NUM_RANKS`	int	1	`{1, 2, 4}`	§3.3	Number of physical ranks (per-rank `CS_n`, `CKE`, `ODT`). 1 = single-rank point-to-point; 2 / 4 = DIMM-class. Rank bits stack at the top of the address map (§3.1) and refresh state is tracked per (rank, bank).
`NUM_BANKS`	int	8	`{4, 8}`	§3.3	Per-device bank count (per rank). Sets the bank-machine / lookup count.
`ROW_WIDTH`	i	14	12-16	§3.	Row-address width.

Parameter	Type	Default	Range / valid set	Section	Purpose
	nt			3	
COL_WIDTH	int	10	9-12	§3.3	Column-address width. Also the ROW_MAJOR setting of ADDR_MAP.bank_lsb.
DFI_RATE	int	2	{1, 2, 4}	§3.6	DFI frequency-ratio gear (phase count). Drives all DFI bus widths and the internal data-unit width. This is the gear parameter formerly called N_PHASES.
DRAM_BEAT_WIDTH	int	64	{16, 32, 64}	§3.6	Width of one DRAM data beat (per DFI phase).
BL	int	8	{4, 8}	§3.6	DRAM burst length in beats. The AXI intake splits host bursts into fixed-BL commands; one AXI burst at the core (DW) side is (awlen+1)*DFI_RATE == BL.
NUM_ENTRIES	int	8	4-32 (power of 2)	§3.2	Depth of the read/write command CAMs (in-flight transaction slots). Pointer width is clog2(NUM_ENTRIES).
N_SRAM_SLOTS	int	8	4-32 (power of 2)	§3.2	Write-data SRAM slot count (buffered write payloads awaiting commit).
BYTE_OFFSET_WIDTH	int	3	clog2(beat byte size)	§3.1	Low byte-offset bits stripped before address decode (log2 of beat byte size).
AGE_WIDTH	int	16	8-24	§3.2	Width of the per-transaction age counter used for FR-FCFS anti-starvation tie-breaking.
HOST_AXI_DATA_WIDT	i	12	{64, 128, 256}	§3.	**pumice_top_geared only.** Free host-

Parameter	Type	Default	Range / valid set	Section	Purpose
H	nt	8	(verified)	1	side AXI data width. Formally-verified axi4_dwidth_converter_wr/_rd shims bridge the host width to the fixed core width DW. HOST_AXI_DATA_WIDTH == DW is a generate-bypassed, bit-identical direct connection (no converter, no added latency).

Derived parameters (computed at elaboration, not overridable independently):

- DW = DRAM_BEAT_WIDTH * DFI_RATE — the fixed core AXI data width **and** the DFI word width (default 64 * 2 = 128). DFI_DATA_WIDTH = DW. One core AXI beat == one DFI word == DFI_RATE DRAM beats.
- SW = DW / 8 — core AXI / DFI strobe width.
- PHW = clog2(DFI_RATE) (min 1) — DFI sub-phase index width.
- DFI_STRB_WIDTH = DW/8, DFI_EN_WIDTH = DFI_VALID_WIDTH = DFI_RATE, DFI_ADDR_BUS_W = ROW_WIDTH*DFI_RATE, DFI_BANK_BUS_W = clog2(NUM_BANKS)*DFI_RATE, DFI_CS_BUS_W = NUM_RANKS*DFI_RATE — the DFI 2.1 pin-bus widths, all scaled by DFI_RATE.

Runtime CSR knobs (not build parameters):

These select behavior at runtime and have no build-time parameter in this generation. Full field detail is in §6.3.

- PHY_TIMING.memtype (1-bit) — 0 = DDR2, 1 = LPDDR2. The core decodes memtype_e from this field (pumice_top.sv w_memtype); there is no build-time MEMTYPE string.
- DFI_PHASE.wr_phase / DFI_PHASE.rd_phase (3-bit each, sliced to PHW downstream) — which DFI sub-phase carries the WRITE / READ command. Defaults 0/0. These replace the former build-time WRPHASE / RDPHASE.
- ADDR_MAP.bank_lsb (5-bit, reset 0x0A = COL_WIDTH = ROW_MAJOR), ADDR_MAP.hash_en (1-bit), ADDR_MAP.hash_seed (8-bit) — the single address-map placement knob (+ optional bank XOR-hash). The classic ROW_MAJOR / BANK_INTERLEAVE / XOR_HASH “schemes” are just settings of these fields; there is no ADDR_MAP_SCHEMES_SYNTH /

ADDR_MAP_SCHEME_DEFAULT build parameter and no scheme mux. See addr_mapper.sv.

- REFRESH_TUNING.page_policy_or (2-bit) — 00 = build-time default, 01 = OPEN, 10 = CLOSE, 11 = HAPPY_HYBRID.
- REFRESH_TUNING.refpb_policy_or (2-bit), REFRESH_TUNING.refresh_defer_active (4-bit), REFRESH_TUNING.zqcs_freq_hz (16-bit, reset 1 Hz).
- SCHED_TUNING.lookahead_active (4-bit), force_inorder (1-bit), happy_enable (1-bit, reset 1), age_max_runtime (8-bit), txn_queue_high_water (8-bit); lookahead_max_obs (4-bit, RO echo of the build MAX).
- PAGE_PRED_TUNING.warmup_cycles (16-bit, reset 1024), hysteresis (8-bit, reset 2).
- Timing CSRs — TIMINGS_RC_RCD_RP_RAS (tRC/tRCD/tRP/tRAS), TIMINGS_RFC_REFI (tRFC/tREFI), TIMINGS_RRD_FAW_WTR_CCD (tRRD/tFAW/tWTR/tCCD), TIMINGS_CL_CWL_WR (CL/CWL/tWR/tRFCpb), TIMINGS_RTP_RTW (tRTP/tRTW).
- PHY_TIMING.refresh_burst (4-bit, 1..8), PHY_TIMING.t_phy_wrlat (8-bit), PHY_TIMING.t_rddata_en (8-bit).
- Init timings — INIT_TIMING0 (t_init_wait/t_dll_wait), INIT_TIMING1 (t_mrd_wait/t_rp_wait/t_rfc_wait), INIT_TUNING.zq_retries (4-bit, reset 3), INIT_TUNING.init_timeout_ms (8-bit, reset 10).

These cost zero silicon area beyond the register flops; making them build-time parameters would have been wrong (and was, in earlier revisions of this document).

23.2 Memtype-Dependent Constraints

MEMTYPE is a runtime CSR field (PHY_TIMING.memtype) rather than an elaboration parameter, so the DDR2-vs-LPDDR2 selection does not gate elaboration-time asserts. The memtype-dependent behavior is instead enforced at run time within the synthesized datapath:

- PHY_TIMING.memtype == 1 (LPDDR2) enables the LPDDR2-only features (PASR bank/segment masks, DPD request, temperature-derate readback, per-bank tRFC via TIMINGS_CL_CWL_WR.tRFCpb).
- PHY_TIMING.memtype == 0 (DDR2) uses the DDR2 command encoding and all-bank refresh.

- The DFI address bus is `ROW_WIDTH * DFI_RATE` wide (`DFI_ADDR_BUS_W`) and carries the LPDDR2 CA packing when `memtype == 1`; it must be wide enough for whichever memtype is programmed, which is guaranteed by `ROW_WIDTH >= 14`.

23.3 Timing-Configuration Sanity Checks

The controller expects the following relationships to hold on the loaded timing CSR values. Because timings are runtime CSRs, these are boot-time programming contracts the SoC firmware must honor (they are documented here so the sweep tool and bring-up scripts can assert them):

- `tREFI_cycles >= 100` — the MC clock must be fast enough relative to tREFI that the refresh timer is well-resolved.
- `tRCD_cycles >= 2`, `tRP_cycles >= 2`, `tRC_cycles >= tRAS + tRP` — basic JEDEC consistency checks.
- `tRFC_cycles > tRP_cycles` — the refresh-cycle time must exceed the precharge time.

These are configuration bugs that should be caught at boot time before traffic starts, not preventable run-time faults.

23.4 Parameter Categories

Category	Parameters
Geometry	`NUM_RANKS`, `NUM_BANKS`, `ROW_WIDTH`, `COL_WIDTH`
Multi-rank	`NUM_RANKS`
Bus widths	`AXI_ID_WIDTH`, `AXI_ADDR_WIDTH`, `DRAM_BEAT_WIDTH`, `BYTE_OFFSET_WIDTH`, `HOST_AXI_DATA_WIDTH`; derived `DW`/`DFI_DATA_WIDTH`
Gear / DFI	`DFI_RATE`, `BL` (with runtime `DFI_PHASE.rd_phase` / `DFI_PHASE.wr_phase`)
Capacity	`NUM_ENTRIES`, `N_SRAM_SLOTS`, `AGE_WIDTH`
Characterization	Runtime CSR knobs paired with the geometry above — `SCHED_TUNING.lookahead_active`, `REFRESH_TUNING.page_policy_or` / `refpb_policy_or` / `refresh_defer_active`, `PAGE_PRED_TUNING.*`, `ADDR_MAP.bank_lsb` / `hash_en` (see §5.3 and §6.3)

24 Characterization Knobs

A subset of the build-time parameters is intentionally left configurable for **characterization sweeping**. These are the algorithmic choices that have research-backed alternatives where the best choice depends on workload. Rather than baking in a one-size-fits-all default, we expose them as parameters so the verification plan can sweep them on representative AXI traffic and pick defaults from data.

24.1 Characterization Parameters

Parameter	Choices	Recommended default	Rationale
LOOKAHEAD_DEPTH	0 / 1 / 2 / 4	1	Same-bank lookahead window for **exact** auto-precharge decisions when the next same-bank request is already in the queue (typical for streaming / DMA). Falls back to PAGE_POLICY when the queue is shallow. 0 disables; 2 or 4 buys diminishing returns at modest comparator cost.
PAGE_POLICY	OPEN / CLOSE / HAPPY_HYBRID	HAPPY_HYBRID (LPDDR2), OPEN (DDR2)	Fallback for when lookahead is inconclusive. Streaming workloads favor OPEN; mobile-mixed favors HAPPY_HYBRID. Ghasempour 2015 reports near-optimal accuracy with small storage.
PAGE_PREDICTOR_TABLE_BITS	8 – 16	12	4 KB predictor is sufficient for typical mobile workloads. Larger tables marginal improvement.
REFPB_POLICY	ROUND_ROBIN / OLDEST_FIRST /	DARP	Chang HPCA 2014: DARP wins over round-robin on uneven bank activity

Parameter	Choices	Recommended default	Rationale
	DARP		(real workloads).
REFRESH_DEFER_MAX	1 – 8	8 (JEDEC ceiling)	Stuecheli MICRO 2010: deferring up to 8 maximizes Elastic Refresh benefit. Set to 1 for real-time predictability.
TXN_QUEUE_DEPTH	4 – 64	16	Larger queues help bandwidth on multi-master AXI; smaller queues reduce area.
AGE_MAX	32 – 1024	256	Caps FR-FCFS starvation; larger values let row-hits dominate longer at the cost of tail latency.
DFI_RATE	1 / 2 / 4	depends on PHY	Build-time DFI gear (phase count). 4 typical for FPGA SoCs; 2 for slower mobile platforms; 1 for simulation-only. Formerly named N_PHASES.
ADDR_MAP.bank_lsb (+ hash_en)	log2(BL/DFI_RATE) .. COL_WIDTH, hash off / on	bank_lsb = COL_WIDTH (ROW_MAJOR), hash off	Runtime address-map placement knob (not a build parameter). bank_lsb = COL_WIDTH = ROW_MAJOR (best for streaming); lowering bank_lsb toward log2(BL/DFI_RATE) interleaves banks (best for random); hash_en folds row bits into the bank index to defeat power-of-two-stride hot-banking (adversarial). Sweep bank_lsb across the legal range with hash off, then repeat the best point with hash on. See addr_mapper.sv.

24.2 Sweep Plan

The verification plan (see §6.4) includes a characterization sweep that runs each parameter through its choices on a small benchmark suite:

24.2.1 Benchmark Suite

A key open question for the sweep is the **interaction between `LOOKAHEAD_DEPTH` and `PAGE_POLICY`**. With deep queues and conclusive lookahead, the choice of `PAGE_POLICY` barely matters because the fallback rarely fires. With shallow queues, `PAGE_POLICY` dominates. The sweep matrix should cover both extremes.

Workload	Pattern
`stream-read`	Sequential 4 KB read bursts
`stream-write`	Sequential 4 KB write bursts
`stream-mixed`	Interleaved sequential read and write
`random-narrow`	Random small AXI bursts (1–4 beats)
`random-wide`	Random large AXI bursts (16–64 beats)
`mobile-mixed`	Bursty traffic with periodic large writes (modeled on smartphone GPU + display)
`multi-master`	Two AXI masters issuing concurrent traffic with different patterns

24.2.2 Metrics Collected

For each (workload, parameter) combination, the sweep records:

- **Sustained read bandwidth** (% peak)
- **Sustained write bandwidth** (% peak)
- **Average read latency**
- **99th-percentile read latency**
- **Refresh blocking time** (% of cycles where the bus was blocked by refresh)
- **Row-hit rate** (per-bank, averaged)
- **Refresh deferral utilization** (avg `refresh_pending` over time)

24.2.3 Default Selection

After the sweep, each parameter's default is set to the value that:

1. Maximizes sustained bandwidth across the benchmark suite
2. Subject to: 99th-percentile latency does not regress by more than 20%
3. Subject to: area cost is within budget

The defaults shown in the parameter table above are the **expected** outcomes based on the cited research. The actual defaults will be confirmed by the sweep.

24.3 Runtime Overrides via CSR

Some characterization parameters are also accessible at runtime via CSR (see §6.3). This allows in-system tuning without rebuild:

- `PAGE_POLICY` runtime override (`REFRESH_TUNING.page_policy_or`)
- `REFPB_POLICY` runtime override (`REFRESH_TUNING.refpb_policy_or`)
- Address-map placement (`ADDR_MAP.bank_lsb`, `hash_en`, `hash_seed`) — this is a runtime knob outright, not an override of a build default
- Page predictor warm-up cycles and hysteresis (`PAGE_PRED_TUNING.warmup_cycles` / `hysteresis`)

Runtime overrides take effect at the next "quiet point" (no commands in flight). The SoC is responsible for ensuring no DRAM traffic during the transition.

24.4 Observability for Sweep

The CSR exposes the following observation counters (described in §6.3):

- `OBS_ROW_HIT[8]` — per-bank rolling row-hit count (clear-on-read).
- `OBS_REF_LATENCY[8]` — per-bank average refresh-blocking cycles.
- `OBS_TXN_QUEUE_DEPTH_MAX` / `OBS_TXN_QUEUE_DEPTH_AVG` — max and time-averaged transaction-queue depth.
- `OBS_PAGE_PRED_ACCURACY` — HAPPY-mode rolling prediction accuracy (%).
- `OBS_REFRESH_DEFER_HIST_0..3` — refresh-deferral histogram bins (plus `OBS_REFRESH_PENDING_MAX` for the max `refresh_pending` observed).
- `OBS_AXI_R_LATENCY_AVG` / `OBS_AXI_R_LATENCY_P99` / `OBS_AXI_W_LATENCY_AVG` — AXI read/write latency counters.

These feed back into the sweep tool for in-loop default selection.

25 Family-Wide Config Bits

This section is the **family-wide config-bit registry** — a catalog of runtime configuration bits used across the DDR2 / LPDDR2, DDR3 / LPDDR3, and DDR4 / LPDDR4 memory controllers. Some bits apply to all generations; some are introduced in one generation and absent in earlier ones; some are *defined* across the family but ignored in flavors where the underlying mechanism does not exist. **That last case is intentional.** Allocating the bit family-wide — even when unused — keeps the CSR map mechanically compatible across controllers and lets software treat the family as a single product line.

25.1 Why a Family-Wide Registry

Each controller in the family has its own CSR map (per §6.3 of each generation's HAS). But the field encodings, bit positions within shared registers, and the *names* are aligned across the family. The intent:

- **Software portability**: a single device driver template understands the bit layout and only needs per-flavor capability checks.
- **Bring-up portability**: bring-up software written for DDR3 should mostly work on DDR2 with the same register reads.
- **Verification portability**: the cocotb test suite's CSR-poke helpers can be reused.
- **Debug consistency**: the same "config dump" script produces a parseable output regardless of which flavor is running.

A bit may be marked **N/A** in a flavor; reads return 0 and writes are ignored (no `pslverr`, just silently ignored — because software portability requires that writing a fully-portable config word does not error on flavors that don't implement every bit).

25.2 Applicability Notation

The Apply column uses this shorthand:

- **all** — applies to every flavor in the family
- **DDR-only** — applies to DDR2/3/4 only; LPDDR variants ignore
- **LP-only** — applies to LPDDR2/3/4 only; DDR variants ignore
- **DDR3+** — applies to DDR3, DDR4, and forward; DDR2 ignores
- **DDR4+** — applies to DDR4 only (introduced for bank groups); earlier ignore
- **LPDDR3+** — LPDDR3 and LPDDR4 only
- **LPDDR4-only** — LPDDR4 only

25.3 SCHED_TUNING (Scheduling Family Bits)

Register block at the same offset (0x040) in every controller. These bits govern the FR-FCFS scheduler's runtime behavior.

Bit Field	Width	Apply	Description
`lookahead_active`	4	all	Runtime lookahead window (0 .. build-time MAX). 0 disables lookahead.
`force_inorder`	1	all	Forces FR-FCFS into first-ready FIFO mode for debug / real-time.
`happy_enable`	1	all	Runtime kill-switch for the HAPPY page predictor. Reads as 0 in flavors built without HAPPY.
`age_max_runtime`	8	all	Anti-starvation age cap; runtime override of `AGE_MAX`.
`txn_queue_high_water`	8	all	Backpressure threshold; AXI `awready` deasserts when queue ≥ threshold.
`lookahead_max_obs`	4	all (R only)	Echo of build-time `LOOKAHEAD_DEPTH_MAX`. Software discovery.
`qos_high_priority`	1	all	**PLANNED — not in the RDL yet** (roadmap Axis 1, QoS step): when 1, scheduler boosts requests with `awqos`/`arqos` ≥ 8.
`bank_group_balance`	1	DDR4+	Round-robin across bank groups in scheduler tie-break (DDR4/LPDDR4 only); N/A elsewhere.

25.4 REFRESH_TUNING (Refresh Family Bits)

Bit Field	Width	Apply	Description
`refpb_policy_or`	2	LP-only	REFpb selection-policy override (LPDDR2/3/4); N/A for DDR-only flavors
`page_policy_or`	2	all	Runtime override for fallback page policy
`refresh_defer_active`	4	all	Active deferral count (1 .. build-time MAX). 1 = no batching.
`zqcs_freq_hz`	16	all	Periodic ZQCS interval. 0 = init-only.

Bit Field	Width	Apply	Description
fgr_mode	2	DDR3+	Fine-Granularity Refresh mode (DDR3 1x/2x/4x; DDR4 fixed-rate, on-the-fly). N/A in DDR2.
refresh_per_rank	1	all	Force per-rank dispatch even when NUM_RANKS=1 (debug). Reset = 1 always when multi-rank.

25.5 ADDR_MAP (Address-Mapping Family Bits)

The address-map register at 0x04C is a single **placement knob**, not a scheme mux. The old ADDR_MAP_TUNING register (with a scheme_or selector and a synth_mask_obs synthesized-scheme bitmask) has been **retired** in this generation: the classic ROW_MAJOR / BANK_INTERLEAVE / XOR_HASH schemes are just settings of bank_lsb plus the optional hash_en fold, so there is nothing to mux (see §5.4 §3/§4 and rtl/fub/addr_mapper.sv).

Bit Field	Width	Apply	Description
bank_lsb	5	all	Bank-field LSB in the word address. = COL_WIDTH -> ROW_MAJOR; lower -> interleave. RTL clamps to [0, COL_WIDTH].
hash_en	1	all	Enable bank XOR-hash fold (bank ^= fold(row) ^ hash_seed) = the old XOR_HASH scheme.
hash_seed	8	all	XOR-hash seed. Runtime seed change without rebuild.

Family note: for DDR4+ the bank-group field is placed by extending the same bank_lsb-relative stack rather than by a separate bg_field_position selector; that generation's RDL documents the exact field split.

25.6 INIT_TUNING (Init Family Bits)

Bit Field	Width	Apply	Description
zq_retries	4	all	ZQ calibration retry count (DDR2 uses OCD; LPDDR2 uses MR10 ZQ).
init_timeout_ms	8	all	Per-step init timeout (ms; scaled by SIM_INIT_SCALE in sim).
wl_retries	4	DDR3+	Write-leveling retry count. N/A in DDR2 (no write-leveling).

Bit Field	Width	Apply	Description
mpr_enable	1	DDR3+	Multi-Purpose Register read-back during init. N/A pre-DDR3.
ca_train_enable	1	LPDDR3+	LPDDR3/4 CA-bus training. N/A in LPDDR2 (no CA training).
cbt_enable	1	LPDDR4-only	Command-Bus-Training. N/A pre-LPDDR4.

25.7 POWER_TUNING (Power-State Family Bits)

Bit Field	Width	Apply	Description
apd_idle_threshold	16	all	Cycles of bank-idleness before APD entry
srf_idle_threshold	24	all	Cycles before Self-Refresh entry
dpd_enable	1	LP-only	Deep-Power-Down enable. N/A for DDR-only flavors.
pasr_active	1	LP-only (R)	Reflects whether any rank has a non-zero PASR mask. R-only.
ckedis_after_sr	1	DDR3+	Drop CKE after SR-entry for sub-mW power floor. N/A in DDR2.
low_freq_mode	1	DDR3+	Reduced-frequency operation hint to PHY. N/A in DDR2.

25.8 ECC_TUNING (Future — Inline ECC)

This block is reserved for future inline-ECC controllers. The bits are **N/A in all current controllers** (DDR2/3/4 and LPDDR2/3/4 v1) because inline ECC is out of scope for the current generation. Future revisions of any flavor that adopt inline ECC will populate this block.

Bit Field	Width	Apply	Description
ecc_enable	1	(reserved)	Enable inline ECC. Reads as 0 in all v1 controllers.
ecc_correct_mode	2	(reserved)	SECDED / Chipkill / off

25.9 RANK_TUNING (Multi-Rank Family Bits)

These are present in any flavor with `NUM_RANKS > 1`. For `NUM_RANKS = 1` builds, the bits read as 0 and writes are silently ignored.

Bit Field	Width	Apply	Description
`rank_enable_mask`	4	all	Per-rank enable; clearing a bit suppresses all commands to that rank. Up to 4 ranks.
`odt_rule_or`	2	all	Runtime override for `ODT_RULE_MULTIRANK` (00 = build-time default; 01 = JEDEC_DDR2; 10 = JEDEC_LPDDR2; 11 = OFF).
`num_ranks_obs`	3	all (R only)	Echo of build-time `NUM_RANKS`. Software discovery.
`cs_assert_cycles`	4	all	Number of cycles to hold `CS_n` asserted per command; tunable for PHY timing margin.

25.10 Quiet-Point Behavior

Runtime config bits take effect at the **next configuration quiet point** — no DRAM commands in flight, no refresh sequence in progress. The register block does not enforce quiet points automatically; SoC firmware is responsible for sequencing writes at a drain point (typically during init or an orchestrated idle window). This generation's RDL does not implement a `config_apply` strobe or a `config_settled` status bit — a future family revision may add one; today the SoC owns the drain.

The quiet-point requirement is family-wide because every flavor has hand-off state between scheduler / refresh / power-state that would be corrupted by mid-flight reconfiguration.

25.11 Bit Discovery via the ID Register

This generation does not implement a dedicated capability vector; software discovers the build via the `ID` register at `0xFF0` (see §6.3): `memtype`, `n_phases` (gear ratio), and `version` are readable there, and `module_id` is the fixed `0xD2` family tag. Echo fields embedded in the tuning registers (`SCHED_TUNING.lookahead_max_obs`, etc.) expose the remaining synthesized ceilings. A future family revision may add a packed capability vector along the lines below:

Bits	Field	Description
0	`cap_happy`	1 = HAPPY predictor synthesized
1	`cap_bg_balance`	1 = bank-group balance scheduler logic present

Bits	Field	Description
		(DDR4+)
2	cap_fgr	1 = Fine-Granularity Refresh supported (DDR3+)
3	cap_pasr	1 = PASR mask is implemented (LP-only)
4	cap_dpd	1 = Deep-Power-Down supported (LP-only)
5	cap_wl	1 = Write-leveling implemented (DDR3+)
6	cap_cbt	1 = CBT implemented (LPDDR4-only)
7	cap_xor_hash	1 = bank XOR-hash implemented
11:8	cap_max_ranks	Build-time NUM_RANKS (echo of geometry)
15:12	cap_n_phases	Build-time gear ratio (DFI_RATE)
19:16	cap_memtype	0 = DDR2, 1 = DDR3, 2 = DDR4, 4 = LPDDR2, 5 = LPDDR3, 6 = LPDDR4

25.12 Per-Flavor Applicability Matrix

Bit	DDR2	DDR3	DDR4	LPDDR2	LPDDR3	LPDDR4
lookahead_active	√	√	√	√	√	√
force_inorder	√	√	√	√	√	√
happy_enable	√	√	√	√	√	√
bank_group_balance	—	—	√	—	—	√
refpb_policy_or	—	—	—	√	√	√
fgr_mode	—	√	√	—	—	—
bg_field_position	—	—	√	—	—	√
wl_retries	—	√	√	—	—	—
ca_train_enable	—	—	—	—	√	√
cbt_enable	—	—	—	—	—	√
dpd_enable	—	—	—	√	√	√
ckedis_after_sr	—	√	√	—	—	—
rank_enable_mask	√	√	√	√	√	√
odt_rule_or	√	√	√	√	√	√
hash_en / hash_seed	√	√	√	√	√	√

The “—” entries are present in the CSR map at the documented offsets but read 0 and ignore writes in that flavor. Software treats these as “soft-N/A” — not an error condition, just an absent feature.

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26 Clocks and Reset

26.1 Clock Domains

The controller uses two independent clocks. The register block is a PeakRDL passthrough `cpuif` clocked by `aclk` — there is no separate management clock.

26.1.1 `aclk`

Controller clock. All host-AXI4 logic, the scheduler, the bank timers, the write/read CAMs, and the register block (`pumice_csr`) run on `aclk`. The controller-side of the command / write-data / read-data streams into the DFI layer is on `aclk`.

Typical frequency: 100 – 200 MHz on FPGA SoCs; higher on silicon.

26.1.2 `dfi_clk`

PHY / DFI-side clock. The DFI 2.1 command bus, write serializer, and read aligner run on `dfi_clk`; the PHY consumes and drives the DFI pin bus on this clock. The gear ratio (`DFI_RATE`) sets how many DRAM phases each `dfi_clk` cycle carries.

`aclk` and `dfi_clk` are asynchronous to each other. The single crossing between them lives inside `pumice_dfi_layer` (see the CDC summary below).

26.2 Reset Signals

Both resets are active-low.

26.2.1 `aresetn` (active low, controller / AXI domain)

Controller reset, tied to the `aclk` domain. The register block resets on `~aresetn` (`pumice_csr .rst(~aresetn)`). Resets:

- Host AXI4 interface state (`pumice_axi4_ifc`)
- Write-data and read-command CAMs

- Command scheduler and bank timers (`pumice_mem_cmd_scheduler`)
- Refresh manager and init sequencer
- Register block (`pumice_csr`) and observation state

26.2.2 `dfi_rstn` (active low, DFI / PHY domain)

DFI-domain reset, tied to the `dfi_clk` domain. Resets the DFI command path, the write serializer, the read aligner, and the DFI-side of the CDC FIFOs inside `pumice_dfi_layer`.

> Note: `init_sequencer` names its ports `mc_clk` / `mc_rst_n` internally, but the top level (`pumice_top` → `pumice_core`) ties these to `aclk` / `aresetn`.

26.3 Reset Sequencing

26.3.1 Cold Boot

1. Both resets asserted (`aresetn`, `dfi_rstn`)
2. Clocks stable
3. `dfi_rstn` deasserted; `aresetn` deasserted (software programs the CSR timings/phases/policy via the register `cpuif`, or defaults are used)
4. The init sequencer asserts `dfi_init_start_o` and waits on `dfi_init_complete_i` (the PHY runs its own DLL-lock / IO training)
5. The init sequencer walks the JEDEC MRS / precharge / refresh sequence (see the Init Sequences chapter)
6. On `init_done_o`, the controller begins servicing AXI traffic

26.3.2 Warm Reset (Clock-Gated)

The SoC can warm-reset the controller without losing DRAM content:

1. SoC requests self-refresh entry (via `powerdown_ctrl`, optional)
2. Controller acknowledges; DRAM is in self-refresh
3. SoC asserts `aresetn`; clocks may be gated
4. SoC deasserts `aresetn`; clocks resume
5. Controller re-runs the init sequence
6. Controller resumes normal operation

26.4 Clock Domain Crossing Summary

There is exactly one clock-domain crossing in the design: the `aclk` ↔️ `dfi_clk` crossing inside `pumice_dfi_layer` (`pumice_dfi_cdc.sv`). It is built from asynchronous `gaxi` FIFOs only —

command, write-data, and read-data each get their own async FIFO, plus the `init_start` / `init_complete` handshake. The register `cpuif` shares the `aclk` domain, so there is no CSR-clock CDC.

Crossing	Mechanism	Latency
`aclk` → `dfi_clk`	Async `gaxi` FIFOs (cmd / wrdata) in `pumice_dfi_cdc`	FIFO + `N_FLOP_CROSS`-flop sync
`dfi_clk` → `aclk`	Async `gaxi` FIFO (rddata) in `pumice_dfi_cdc`	FIFO + `N_FLOP_CROSS`-flop sync

26.5 Reset and Power Considerations

DFI v2.1 requires `dfi_init_start` to be asserted by the controller as part of init; the init sequencer drives `dfi_init_start_o` out of reset (once it leaves `S_RESET`) and waits on the PHY's `dfi_init_complete_i` before proceeding to the MRS / precharge / refresh walk. The `init_start` / `init_complete` handshake crosses the `aclk` ↔ `dfi_clk` boundary through the same CDC as the command and data streams.

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27 Init Sequences

The cold-boot init sequences for DDR2 and LPDDR2 are implemented as a single FSM in `rtl/fub/init_sequencer.sv` (not separate step-package files). The FSM issues real MRS / precharge / refresh commands to the DRAM through the command scheduler while `init_busy_o` is high, and updates the mode-register shadow in lockstep so the live CL / CWL / BL decode tracks what was programmed. The `memtype_i` input selects the DDR2 or the LPDDR2 branch.

The DDR2 branch mirrors LiteDRAM's `get_ddr2_phy_init_sequence` and is proven on the Nexys A7 board.

27.1 DDR2 Cold Init

Per JESD79-2F. FSM states from `init_sequencer.sv`; MR data rides the ROW request field (`init_cmd_row_o`), and the MR / EMR number is carried in the bank index (`init_cmd_bank_o`).

Step	State	Command	MR data	Post-wait
1	S_DFI_INIT	assert dfi_init_start, wait dfi_init_complete, then tINIT	—	t_init_wait
2	S_PREA1	PRECHARGE all	—	t_rp_wait (tRP)
3	S_EMR2	MRS EMR(2)	0x0000	t_mrd_wait (tMRD)
4	S_EMR3	MRS EMR(3)	0x0000	t_mrd_wait
5	S_EMR1	MRS EMR(1)	0x0000	t_mrd_wait
6	S_MR0_DLL	MRS MR0 + DLL reset	MR0.VAL | 0x100 (reset default 0x0533 = BL8 / CL3 / tWR3 / DLL_RESET)	t_dll_wait (tDLLK)
7	S_PREA2	PRECHARGE all	—	t_rp_wait
8	S_REF1	AUTO REFRESH	—	t_rfc_wait (tRFC)
9	S_REF2	AUTO REFRESH	—	t_rfc_wait
10	S_MR0	MRS MR0 (DLL reset cleared)	MR0.VAL (reset default 0x0433)	t_mrd_wait
11	S_OCD_DEF	MRS EMR(1) + OCD default	0x0380	t_mrd_wait
12	S_OCD_EXIT	MRS EMR(1) + OCD exit	0x0000	t_mrd_wait
13	S_DONE	assert init_done	—	—

Note the JEDEC extended-mode-register order: EMR(2), then EMR(3), then EMR(1), before MR0. The OCD-default state (S_OCD_DEF) leaves the MR1 shadow unchanged; S_OCD_EXIT restores it to the final 0x0000 value so the live decode is stable.

27.2 LPDDR2 Cold Init

Per JESD209-2F. LPDDR2 uses the Mode Register Write (MRW) chain. Because the MR address (MA) can reach MR63 / MR10 — beyond a 3-bit bank port — the sequencer packs `{MA[5:0], OP[7:0]}` into the ROW request field, and `dfi_cmd_formatter.sv` unpacks it (row[13:8] = MA, row[7:0] = OP) to build the bit-exact LPDDR2 CA-bus MRW word. Only MR1 / MR2 / MR3 update the CL / CWL / BL decode shadow; MR63 and MR10 are issued to the DRAM but not shadowed.

Step	State	Command	OP data	Post-wait
1	`S_DFI_INIT`	assert `dfi_init_start`, wait `dfi_init_complete`, then tINIT	—	`t_init_wait`
2	`S_L_RESET`	MRW(MR63) Reset	`0x00`	`t_init_wait` (tINIT4)
3	`S_L_ZQ`	MRW(MR10) ZQ Init Calibration	`0xFF`	`t_dll_wait` (tZQINIT)
4	`S_L_MR1`	MRW(MR1) BL8 / nWR3	`0x23`	`t_mrd_wait` (tMRW)
5	`S_L_MR2`	MRW(MR2) RL3 / WL1	`0x01`	`t_mrd_wait`
6	`S_L_MR3`	MRW(MR3) DS 40 ohm	`0x02`	`t_mrd_wait`
7	`S_DONE`	assert `init_done`	—	—

The LPDDR2 branch reuses the same CSR waits as DDR2: `t_init_wait` covers both tINIT and tINIT4, `t_dll_wait` covers tZQINIT, and `t_mrd_wait` covers the post-MRW tMRW delay.

LPDDR2 is now fully functional in sim: reads and writes, bit-exact JESD209-2F CA encoding, and the full MR init chain above.

27.3 Inter-Command Waits (CSR-Backed)

The JEDEC inter-command delays were previously hardcoded; they are now driven by CSR fields (zero-extended to the 16-bit internal countdown). All counts are in MC (`aclk`) cycles.

CSR field	Purpose	Reset value
`INIT_TIMING0.t_init_wait`	CKE / tINIT settle	512
`INIT_TIMING0.t_dll_wait`	DLL lock (tDLLK)	256
`INIT_TIMING1.t_mrd_wait`	post mode-register (tMRD)	8
`INIT_TIMING1.t_rp_wait`	post precharge (tRP)	8

CSR field	Purpose	Reset value
`INIT_TIMING1.t_rfc_wait`	post auto-refresh (tRFC)	16

Init is a one-time event, so generous margins are fine — the reset defaults comfortably cover the JEDEC minimums at the on-board clock rate. Simulation programs shorter `INIT_TIMING*` values to keep test runtimes practical; there is no separate scaling parameter — the shorter waits are just smaller CSR values.

27.4 Power-State Transitions

Self-refresh and deep-power-down transitions are handled by the optional `powerdown_ctrl.sv` power-state FSM, not by the init sequencer. On an idle threshold it requests a low-power state and drops CKE (`dfi_cke_o`); it wakes on any new controller activity. The two supported entry modes are:

- Precharge Power Down (PDE) — CKE low, banks may stay active.
- Self Refresh (SR) — CKE low plus the SRE command; the DRAM self-refreshes internally and the controller stops issuing REFs.

Self-refresh entry requires all banks precharged first (the scheduler enforces this precondition before granting). Self-refresh exit timing (tXSDLL for DDR2 / tXSR for LPDDR2) is enforced by the scheduler; the power-state FSM itself only drops and restores CKE.

LPDDR2 Deep Power Down (DPD) is supported as a planned power-state entry (CKE low after precharge-all). DPD exit loses DRAM content, so the full LPDDR2 cold init sequence above must run again.

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28 CSR Register Map

The controller's configuration and observation registers are defined in `rtl/macro/pumice_csr.rdl` and generated with PeakRDL into a passthrough-cpuif register block (`bin/peakrdl_generate.py`). The DV mirror is `dv/tbclasses/pumice_regmap.py`; software addresses every field **by name** through that generated map — hardcoded offsets are forbidden. This chapter is the authoritative narrative for that RDL; every register, offset, and reset value below is reconciled against it.

The map occupies a 4 KB space (12-bit address, `CSR_ADDR_W = 12`). All registers are 32 bits wide, little-endian, 4-byte aligned. The register bus is the PeakRDL **passthrough cpuif** (`s_cpuif_*`), not a hand-written APB slave — see §4.3. Config fields are consumed `hw = r` (`hwif_out.*`) and drive the core by name; status/observation fields are `hw = w` (`hwif_in.*`).

28.1 Control and Status (0x000 – 0x00F)

28.1.1 CTRL (0x000, R/W)

Bit	Field	Reset	Description
0	`init_start`	0	Write 1 to start init (self-modifying strobe)
1	`init_force_restart`	0	Write 1 to force re-init even mid-sequence
3:2	`RSVD_3_2`	0	Reserved
4	`pwr_req_low_power`	0	Request power-down state
5	`pwr_req_dpd`	0	Request DPD (LPDDR2 only)
6	`pwr_req_active`	0	Request return to ACTIVE
7	`pwr_req_self_refresh`	0	Request self-refresh
30:8	`RSVD_30_8`	0	Reserved
31	`soft_reset`	0	Write 1 to assert internal soft reset (self-clearing)

28.1.2 STATUS (0x004, R only)

Bit	Field	Description
0	`init_done`	1 when init complete
1	`init_error`	1 on init failure
7:4	`power_state`	Current power-state FSM state (encoded)
8	`pasr_active`	LPDDR2: PASR mask is non-zero
23:16	`init_step_dbg`	Current init step number (for bring-up)
31	`version_match`	1 when build matches expected version

28.1.3 STATUS_HISTORY (0x008, R only)

Last 8 power-state transitions, 4 bits each. Most recent in `[3:0]`; new transitions push the oldest off. Useful for bring-up debugging of power-state oscillation.

28.2 Timing Parameters (0x010 – 0x01F)

Packed timing parameters in MC (`aclk`) cycles. All fields are `sw = rw`, `hw = r` and drive the core scheduler / DFI layer.

28.2.1 TIMINGS_RC_RCD_RP_RAS (0x010, R/W)

Bits	Field	Reset	Description
7:0	tRC	60	tRC in cycles
15:8	tRCD	15	tRCD in cycles
23:16	tRP	15	tRP in cycles
31:24	tRAS	40	tRAS in cycles

28.2.2 TIMINGS_RFC_REFI (0x014, R/W)

Bits	Field	Reset	Description
15:0	tRFC	16	tRFC (or tRFCab) in MC cycles — mission-mode REF recovery, consumed by the arbiter
31:16	tREFI	1950	tREFI in cycles

28.2.3 TIMINGS_RRD_FAW_WTR_CCD (0x018, R/W)

Bits	Field	Reset	Description
7:0	tRRD	6	tRRD
15:8	tFAW	35	tFAW
23:16	tWTR	4	tWTR
31:24	tCCD	4	tCCD

28.2.4 TIMINGS_CL_CWL_WR (0x01C, R/W)

Bits	Field	Reset	Description
7:0	CL	6	CAS latency
15:8	CWL	4	CAS write latency

Bits	Field	Reset	Description
23:16	tWR	15	Write recovery
31:24	tRFCpb	70	LPDDR2 per-bank tRFC

28.3 Mode Register Values (0x020 – 0x02F)

28.3.1 MR0, MR1, MR2, MR3 (0x020, 0x024, 0x028, 0x02C, R/W)

Low 16 bits (VAL) contain the MR value; upper 16 bits reserved. These hold override MR values; the default init walk uses the JEDEC-standard values built into `init_sequencer.sv` (§6.2). Reset = 0.

28.4 LPDDR2-Specific (0x030 – 0x03F)

28.4.1 PASR_BANK_MASK_RANK0 (0x030, R/W)

Bits	Field	Reset	Description
7:0	pasr_banks	0	LPDDR2 PASR per-bank mask for rank 0 (MR16); bit N = 1 masks bank N

28.4.2 PASR_SEG_MASK_RANK0 (0x034, R/W)

Bits	Field	Reset	Description
7:0	pasr_segs	0	LPDDR2 PASR segment mask for rank 0

28.4.3 TEMP_DERATE_RANK0 (0x038, R only)

Bits	Field	Description
1:0	temp_class	From rank 0's MR4: 00 = nominal, 01 = 2x refresh, 10 = 4x refresh

The current RDL defines the single-rank instances only (NUM_BANKS = 8, rank 0). Multi-rank builds would extend this block; the map as generated today does not allocate per-rank PASR/temperature registers beyond rank 0.

28.5 Scheduler / Refresh / Page / Addr / Init Tuning (0x040 – 0x05F)

28.5.1 SCHED_TUNING (0x040, R/W)

Bits	Field	Reset	Description
3:0	lookahead_active	0	Active lookahead window (0..LOOKAHEAD_DEPTH_MAX). 0 disables lookahead.
4	force_inorder	0	1 = force first-ready FIFO (disable row-hit reordering).
5	happy_enable	1	1 = HAPPY predictor active (only meaningful if synthesized).
15:8	age_max_runtime	0	Runtime AGE_MAX override (0 = use build-time default).
23:16	txn_queue_high_water	0	Backpressure-assertion threshold for the txn queue.
27:24	lookahead_max_obs	—	(R only, hw = w) Echo of build-time LOOKAHEAD_DEPTH_MAX.

28.5.2 PAGE_PRED_TUNING (0x044, R/W)

Bits	Field	Reset	Description
15:0	warmup_cycles	1024	Warmup cycles before predictor is trusted
23:16	hysteresis	2	Saturating-counter hysteresis

28.5.3 REFRESH_TUNING (0x048, R/W)

Bits	Field	Reset	Description
1:0	refpb_policy_or	0	00 = build-time; 01 = RR; 10 = OLDEST_FIRST; 11 = DARP.
3:2	page_policy_or	0	00 = build-time; 01 = OPEN; 10 = CLOSE; 11 = HAPPY_HYBRID.
7:4	refresh_defer_active	1	Active refresh deferral count (1..REFRESH_DEFER_MAX). 1 = no batching.
31	zqcs_freq_hz	1	Periodic ZQCS interval in Hz. 0 disables periodic

Bits	Field	Reset	Description
:16			ZQCS.

28.5.4 ADDR_MAP (0x04C, R/W)

Replaces the retired `ADDR_MAP_TUNING` register. The AXI-to-DRAM mapping is driven by a single knob rather than a scheme selector — see §5.4 and §3.1, and `rtl/fub/addr_mapper.sv`.

Bits	Field	Reset	Description
4:0	`bank_lsb`	0x0A	Bank-field LSB in the word address. Default 10 = `COL_WIDTH` = ROW_MAJOR placement. RTL clamps to `[0, COL_WIDTH]`.
8	`hash_en`	0	Enable bank XOR-hash: `bank ^= fold(row) ^ hash_seed`.
23:16	`hash_seed`	0	XOR-hash seed (`seed[BW-1:0]`).

The register reset value is `0x0000000A` (`bank_lsb = 10`, hash off). Setting `bank_lsb = COL_WIDTH` gives ROW_MAJOR; `bank_lsb = log2(cols/burst)` gives maximum bank interleave with burst locality preserved; `hash_en` folds an XOR-hash on top of any placement (the old XOR_HASH scheme). There is no separate scheme-selector field.

28.5.5 INIT_TUNING (0x050, R/W)

Bits	Field	Reset	Description
3:0	`zq_retries`	3	ZQ calibration retry count before raising `init_error`. Range 1–8.
15:8	`init_timeout_ms`	10	Per-step init timeout in ms. Range 1–255.

28.5.6 INIT_TIMING0 (0x058, R/W)

JEDEC init-sequence waits (MC cycles), consumed by `init_sequencer.sv`. Previously hardcoded.

Bits	Field	Reset	Description
15:0	`t_init_wait`	512	CKE / tINIT settle
31:16	`t_dll_wait`	256	DLL lock

Bits	Field	Reset	Description
			(tDLLK)

28.5.7 INIT_TIMING1 (0x05C, R/W)

Bits	Field	Reset	Description
7:0	t_mrd_wait	8	tMRD (post mode-reg)
15:8	t_rp_wait	8	tRP (post precharge)
23:16	t_rfc_wait	16	tRFC (post refresh)

28.5.8 TIMINGS_RTP_RTW (0x054, R/W)

Read-to-precharge and read-to-write turn-around. Previously tRTP was hardcoded (8'd4) and tRTW was tied to it; both are now independent configs.

Bits	Field	Reset	Description
7:0	tRTP	4	Read to precharge
15:8	tRTW	6	Read to write

28.6 DFI / PHY Timing (0x060 – 0x067)

28.6.1 DFI_PHASE (0x060, R/W)

Which DFI sub-phase carries the READ vs WRITE command, to match the PHY's rdphase / wrphase contract (see §4.2). Fields are sliced to clog2(DFI_RATE) bits downstream; upper bits are ignored when DFI_RATE is narrower than the field. Defaults 0/0 preserve the legacy all-on-phase-0 behavior.

Bits	Field	Reset	Description
2:0	rd_phase	0	READ command DFI sub-phase
6:4	wr_phase	0	WRITE command DFI sub-phase

28.6.2 PHY_TIMING (0x064, R/W)

PHY / DFI data timing plus memory-type selection. All fields `hw = r` so they drive the controller core.

Bits	Field	Reset	Description
7:0	`t_phy_wrlat`	0	WRITE command -> `dfi_wrdata_en` (Nexys A7 bring-up tuple programs 1)
15:8	`t_rddata_en`	6	RD command -> `dfi_rddata_en` window
16	`memtype`	0	0 = DDR2, 1 = LPDDR2 (drives the whole core `memtype_e` path)
23:20	`refresh_burst`	1	REFs drained per request (1..8)

`PHY_TIMING.memtype` is the runtime memory-type select. It is decoded into the `memtype_e` enum at the top and fanned out to the addr/scheduler/DFI/init logic; geometry (DFI_RATE, DRAM_BEAT_WIDTH, NUM_BANKS, ROW/COL width, BL) remains build-time — see §5.1.

28.7 Per-Bank Observation (0x080 – 0x0DF)

Single-rank (rank 0), `NUM_BANKS = 8`. Each is a `regfile` of 8 word-wide registers striding by 4 bytes.

28.7.1 OBS_ROW_HIT[0..7] (0x080 + N×4, R/W with clear-on-read)

Rolling row-hit count per bank. `VAL[31:0]`, `onread = rclr` (reset on read or `soft_reset`).

28.7.2 OBS_REF_LATENCY[0..7] (0x0C0 + N×4, R only)

Average refresh-blocking cycles per bank. `VAL[31:0]`, `hw = w`.

28.8 System Observation (0x100 – 0x1FF)

All `hw = w`, `sw = r`.

Offset	Register	Description
0x100	`OBS_TXN_QUEUE_DEPTH_MAX`	Max queue depth observed
0x104	`OBS_TXN_QUEUE_DEPTH_AVG`	Time-averaged depth

Offset	Register	Description
0x108	OBS_REFRESH_PENDING_MAX	Max refresh_pending value observed
0x10C	OBS_REFRESH_DEFER_HIST_0	Refresh deferral histogram bin 0
0x110	OBS_REFRESH_DEFER_HIST_1	Histogram bin 1
0x114	OBS_REFRESH_DEFER_HIST_2	Histogram bin 2
0x118	OBS_REFRESH_DEFER_HIST_3	Histogram bin 3
0x120	OBS_PAGE_PRED_ACCURACY	HAPPY mode: rolling prediction accuracy (%)
0x130	OBS_AXI_R_LATENCY_AVG	Avg AXI read latency in cycles
0x134	OBS_AXI_R_LATENCY_P99	99th-percentile read latency
0x138	OBS_AXI_W_LATENCY_AVG	Avg AXI write latency

28.8.1 OBS_WORDS[0..8] (0x1C0 + N×4, R only)

Nine 32-bit read-only words carrying the packed obs_* signal harvest from the FUB internals (see docs/csr_obs_layout.md). VAL[31:0], hw = w.

28.9 Module Identification (0xFF0 – 0xFFC)

28.9.1 ID (0xFF0, R only, reset = 0xD2020001)

Bits	Field	Reset	Description
7:0	version	0x01	Build version
15:8	memtype	0x00	0 = DDR2, 1 = LPDDR2
23:16	n_phases	0x02	Gear ratio (1, 2, or 4)
31:24	module_id	0xD2	Fixed 0xD2

28.9.2 BUILD (0xFF4, R only)

Build hash word. `VAL[31:0]`, reset = 0.

28.10 Notes on This Revision

- `ADDR_MAP_TUNING` (with `scheme_or` / `synth_mask_obs`) has been **retired** and replaced by `ADDR_MAP` at 0x04C. No scheme-selector field exists in RTL.
- `TIMINGS_RTP_RTW` (0x054), `INIT_TIMING0` (0x058), `INIT_TIMING1` (0x05C), `DFI_PHASE` (0x060), and `PHY_TIMING` (0x064) expose knobs that were formerly hardcoded in the FUBs; they are now runtime CSRs.
- `PHY_TIMING.memtype` and `PHY_TIMING.refresh_burst` are new runtime fields.
- The observation block is single-rank as generated; the multi-rank per-rank layout described in earlier revisions is not present in the current RDL.

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29 Verification Strategy

A layered verification approach with the existing DFI BFM as the primary reference.

29.1 Layered Approach

29.1.1 Layer 1: Module-Level Unit Tests

Each live FUB gets its own cocotb unit test (see `dv/tests/fub/`):

Module	Test focus
`pumice_wr_intake`	AW-meta + wr-data FIFO handshakes; backpressure; burst splitting into fixed-BL commands
`pumice_rd_intake`	AR channel handshakes; address mapping; snarf (read-your-write) probe
`pumice_wr_data_cam`	WR CAM fill / commit-drain / snarf movers; `r_fdone` fill-complete gating

Module	Test focus
pumice_rd_cmd_cam	RD CAM return-fill / drain movers; oldest-first, data-ready gating
addr_mapper	Address translation across ADDR_MAP.bank_lsb settings (ROW_MAJOR .. interleave) and hash on/off
pumice_cmd_arbiter	Single-pick priority order; inline open-page decision; refresh prioritization
pumice_bank_timers / bank_timer	Per-(rank,bank) preset/decrement timers; combinational safe_act/rd/wr/pre readiness
global_timers	tFAW / tRRD / tWTR / tRTW / tCCD against JESD79-2 / JESD209-2 truth tables
refresh_ctrl	tREFI counter; refresh-deferral; refresh request/ack handshake
init_sequencer	Step-by-step DDR2 and LPDDR2 JEDEC MR init sequences
mode_register	CL / CWL / BL / AL decode (DDR2 and LPDDR2 MR2 RL/WL enum)
dfi_cmd_formatter	Round-trip table tests against JEDEC encoding; LPDDR2 CA-bus branch
dfi_signal_pack	DW-to-phase split; idle-phase handling
pumice_dfi_cdc	The single async-FIFO clock-domain crossing; bubble-free cmd/wr/rd flow
pumice_dfi_wr_serializer	CWL alignment; write-data serialization onto DFI phases
pumice_dfi_rd_aligner	CL alignment; read-data assembly; back-pressure

The four macro wrappers and the integration levels are exercised at Layer 2/3: pumice_axi4_ifc, pumice_mem_cmd_scheduler, pumice_dfi_layer, pumice_core, and pumice_top (the PeakRDL pumice_csr register block is verified with the top).

29.1.2 Layer 2: Subsystem Tests

Multi-module integration tests, framed around the three macro layers (see dv/tests/macro/):

- **AXI interface** (`pumice_axi4_ifc` = wr/rd intakes + the two CAMs): full AXI transaction flow with diverse traffic, including read-your-write snarf forwarding
- **Command scheduler** (`pumice_mem_cmd_scheduler` = cmd arbiter + bank/global timers + refresh + init + mode register): scheduler decisions on representative queue contents, inline open-page reordering, refresh interleave
- **DFI layer** (`pumice_dfi_layer` = single CDC + cmd path + wr serializer + rd aligner): command/data flow across the clock-domain crossing with the DFI BFM slave

29.1.3 Layer 3: End-to-End with DFI BFM Slave

The full controller drives the DFI BFM slave through our existing co-sim harness (`tests/sim/dfi/test_litedram/...` patterns in the DV repo).

- AXI4 stimulus generators issue representative traffic
- DFI BFM slave captures the controller's command stream
- Assertions check: command sequencing matches AXI, timing constraints are honored, data integrity round-trips through the slave's MemoryModel

29.1.4 Layer 4: External-Reference Cross-Validation

For DDR2: cross-validate against LiteDRAM's DDR2 model.

- Generate LiteDRAM controller alongside ours
- Both drive the same DFI BFM slave with the same AXI stimulus
- Compare command streams; differences flag potential design issues

This is a real, proven path. A LiteDRAM DDR2 memtest passes on the Nexys A7 board (128 MiB), which confirmed the board, PHY, pin-out, and DRAM are good. pumice itself is now board-proven on the same Nexys A7: DDR2 reads and writes are clean (0 mismatches across the characterization traffic). Both memtypes pass the full sim suite.

LPDDR2 has no LiteDRAM analog, so cross-validation for LPDDR2 is limited to the BFM master to BFM slave path. LPDDR2 is nonetheless fully functional in sim (reads and writes, bit-exact JESD209-2F CA encoding, full MR init) and passes the sim suite.

29.2 Characterization Sweep

Post-functional verification, the sweep runs each characterization parameter through its choices on the benchmark suite from §5.2:

29.2.1 Sweep Matrix

Parameter	Sweep values
LOOKAHEAD_DEPTH	0, 1, 2, 4
PAGE_POLICY	OPEN, CLOSE, HAPPY_HYBRID
REFPB_POLICY	ROUND_ROBIN, OLDEST_FIRST, DARP
REFRESH_DEFER_MAX	1, 2, 4, 8
TXN_QUEUE_DEPTH	8, 16, 32
DFI_RATE	(build-determined; usually 4)
ADDR_MAP.bank_lsb	ROW_MAJOR (== COL_WIDTH), interleave, hash on/off

The full cross-product would be 4 × 3 × 3 × 4 × 3 = 432 builds; in practice we sweep one parameter at a time holding others at recommended defaults, which reduces to ~17 builds per memtype. The exception is the LOOKAHEAD_DEPTH × PAGE_POLICY pair — these interact strongly (the fallback policy only matters when lookahead is inconclusive), so this pair is swept as a full 4 × 3 = 12-point grid.

29.2.2 Sweep Outputs

Each (parameter, value) run produces a JSON report:

```
{
  "memtype": "LPDDR2",
  "page_policy": "HAPPY_HYBRID",
  "refpb_policy": "DARP",
  "workload": "mobile-mixed",
  "metrics": {
    "read_bw_pct": 73.2,
    "write_bw_pct": 68.1,
    "read_latency_avg_ns": 142,
    "read_latency_p99_ns": 198,
    "refresh_block_pct": 4.3,
    "row_hit_rate_avg": 0.81,
    "page_predictor_accuracy": 0.91
  }
}
```

These feed into a sweep visualization tool that selects defaults from data.

29.2.3 Out-of-Order Scheduler Trade-Offs

The centralized FR-FCFS scheduler proposed in §1 (Differentiating Features) is the most architecturally aggressive choice in this controller and warrants explicit characterization beyond just the sweep matrix above. Three concrete concerns:

- **Timing-critical path scales with `TXN_QUEUE_DEPTH`.** Scanning the full queue every cycle for the best candidate is a comparator pyramid: priority resolves over (ready-flag, row-hit-flag, age) for every entry simultaneously. At `TXN_QUEUE_DEPTH = 16` this is well within an embedded-SoC clock budget; at 32 it stays comfortable; at 64 it becomes the critical path on the controller side. The characterization sweep should report scheduler combinational depth at synthesis time alongside the performance numbers — the right `TXN_QUEUE_DEPTH` is the largest value that doesn't push the scheduler off the clock budget.
- **Area cost is super-linear in queue depth.** Each queue entry needs its full metadata replicated (bank, row, col, ID, age, row-hit cache). At depth 16, ~2K gates; at depth 64, ~10K gates — comparable to the rest of the scheduler combined. The HAPPY predictor's gate cost is a separate line item.
- **Verification cost is significantly higher than FIFO.** OoO issue means assertions like "request N landed at cycle X" can't be used. The DV plan relies on protocol-level scoreboarding: the BFM master records arrival order per ID, the BFM slave records issue order, and the scoreboard verifies the AXI4 per-ID ordering invariant is preserved. Within-ID ordering violations are a higher-severity bug than performance regressions and should be caught at Layer 1 unit tests, not deferred to integration.
- **Fairness tail matters under adversarial workloads.** With `AGE_MAX` as the only anti-starvation cap, pathological patterns can drive 99th-percentile latency well above mean. The `random-narrow` and `multi-master` workloads in the benchmark suite intentionally exercise this; the `mobile-mixed` workload simulates the common case. The recommended default for `AGE_MAX` should be picked based on tail-latency targets, not throughput targets.

These concerns are unique to the OoO scheduler design choice — a simpler per-bank FIFO controller (the typical open-source pattern) would not face them, at the cost of leaving row-hit reordering performance on the table.

29.3 Verification Hooks

Internal observation outputs are exposed via CSR (see §6.3) for in-system characterization. Additionally, the controller emits internal "event" signals that can be tied to waveform-dump triggers:

- `ev_init_done`
- `ev_refresh_issued`
- `ev_page_conflict_detected`
- `ev_queue_full`
- `ev_refresh_pending_critical`

Specific events can be wired to a test-bench-controlled waveform-capture trigger during bring-up.

29.4 Coverage Targets

Functional coverage (collected during regression):

- All FSM states reached in each module that has one (the bank timers and CAMs are FSM-free; their timer/counter states are covered instead)
- All page policies exercised
- All refresh policies exercised
- All scheduler priority levels exercised
- All AXI burst types exercised (if enabled)
- All `ADDR_MAP.bank_lsb` settings (ROW_MAJOR .. interleave) exercised, with hash on and off

Code coverage targets: 100% line / branch coverage for all modules.

Assertion coverage: all design assertions (registered protocol checks) hit during regression.

---

30 Open Issues and TODOs

Items where design choices were made provisionally and should be reviewed before the HAS is promoted to formal status.

30.1 Open Issues

30.1.1 1. AXI4 Narrow-Burst Handling Confirmation

AXI4 allows narrow transfers (data width less than bus width). The current plan is to handle them via byte-mask through `wrdata_mask`. **Confirm** the wstrb-to-wrdata-mask wiring honors the AXI partial-write semantics correctly, particularly for unaligned bursts.

30.1.2 2. AXI4 Read-Modify-Write Strobe Handling

Partial writes in DDR2 / LPDDR2 use `wrdata_mask` to mask byte lanes; this is straightforward. **Confirm** that per-byte masking is wired all the way through the write path (`pumice_wr_intake` -> `pumice_wr_data_cam` -> `pumice_dfi_wr_serializer`) and that `pumice_wr_intake` doesn't widen narrow writes implicitly.

30.1.3 3. CSR Address Space Allocation

The config bus is a PeakRDL passthrough cpuif on `aclk` (no APB slave, no separate `apb_pclk`). `CSR_ADDR_W = 12` provides a 4 KB register region. **Confirm** the SoC's address-map allocation has at least 4 KB available for the cpuif window, and that there's no conflict with adjacent peripherals.

30.1.4 4. Per-Bank Refresh Book-Keeping Cost

Keeping `last_ref_age` per bank costs one counter per bank. For 8 banks this is 8 small counters. **Confirm** the synthesis area impact is acceptable; we expect ~200 LUTs for the per-bank refresh tracking on a typical FPGA.

30.1.5 5. HAPPY Predictor Hash Function

Ghasempour 2015 suggests bank-XOR-low-row-bits as the hash function. **Confirm** with our typical workloads. If the workloads have systematic bank-row correlations, a different hash may be needed.

30.1.6 6. Self-Refresh Exit Latency

JESD79-2 requires `tXSNR` (~200 ns) before any command after SR exit. **Confirm** the optional `powerdown_ctrl` enforces this correctly and that the scheduler is gated for the full duration.

30.1.7 7. Init Sequencer Retry on Error

If ZQ fails, do we retry up to 3 times then halt, or just halt? Current plan: 3 retries (parameter `INIT_ZQ_RETRIES`) then raise `init_error`. **Confirm** with the system architect — some systems prefer fail-fast. (Note: DDR2 ZQ init differs from LPDDR2 — `init_sequencer` handles both JEDEC sequences.)

30.1.8 8. CSR Write Atomicity

Multi-register parameter changes (e.g., updating both `MR0` and a related timing) need a "config quiet period" to take effect atomically. Today the SoC owns the drain: software is expected to quiesce AXI traffic before reprogramming timing/mode CSRs. A dedicated `config_apply` strobe (hardware-enforced quiet period) is **not implemented** and is a possible future addition. **Document** the software drain protocol regardless.

30.1.9 9. Burst Splitting at Row Boundaries

AXI4 allows up to 256 beats per burst; a DRAM row contains `2^COL_WIDTH` columns. Burst splitting into fixed-BL commands happens in `pumice_wr_intake` / `pumice_rd_intake`. **Confirm** the split logic correctly tracks partial-burst completion for ID-based ordering.

30.1.10 10. Multi-Rank

Multi-rank scaffolding now exists: `NUM_RANKS` is a real build parameter (default 1), the DFI `dfi_cs_n_o` / `dfi_odt_o` buses are per-rank (width `NUM_RANKS * DFI_RATE`), and `addr_mapper` carries a rank field (`rank_o`) at the top of the address stack. The command path and bank timers are stamped per (rank, bank). **Remaining gap**: per-rank refresh coordination, PASR (partial-array self-refresh), and per-rank observation registers are not yet implemented. **Confirm** the multi-rank data path with a `NUM_RANKS > 1` build before relying on it.

30.1.11 11. ECC

Out of scope for this controller. ECC is expected to be handled at the SoC level or in a sideband wrapper, not the controller itself.

30.1.12 12. DFI Training Sub-Interface

Not driven in v1; the assumption is that the PHY handles its own training or training is not required for this DDR2 / LPDDR2 target. **Confirm** with the PHY vendor before silicon tape-out.

30.1.13 13. AXI Quality of Service (QoS)

Currently the `awqos` / `arqos` fields are forwarded to the scheduler as priority hints, but the scheduler does not implement explicit QoS classes. **Decide** whether to add proper QoS support (multi-class scheduler) or leave QoS as a future enhancement.

30.1.14 14. Multi-Master AXI

Currently single AXI port. **Decide** whether to add a multi-port AXI crossbar internally (simplifies SoC integration but adds area) or leave as a single-port with the SoC responsible for AXI arbitration upstream.

30.1.15 15. Observation Counter Overflow

Observation counters (row-hit, queue depth max, etc.) are 32-bit. At high traffic rates they could wrap. **Decide** on overflow handling: saturate, clear-on-read, or rely on SoC reading frequently.

30.1.16 16. Retired-Architecture Modules Still on Disk (0LD/)

The pre-rearchitecture modules still live under `rtl/fub/OLD/`, `rtl/macro/OLD/`, and `rtl/top/OLD/` (e.g. `scheduler.sv`, `wr_cmd_cam.sv`, `rd_cmd_cam.sv`, `xbank_timers.sv`, `axi_intake.sv`, the `*_macro.sv` blocks, `pumice_csr_slave.sv`). They are referenced only by retired sentinel tests in `dv/tests/` (`test_scheduler.py`, `test_wr_cmd_cam.py`, `test_xbank_timers.py`, `test_axi_intake.py`, and the `*_macro` tests). **Remove** the `OLD/` trees and their sentinel tests once the new architecture is fully signed off, so the live module set is the only thing that builds.

30.1.17 17. Open-Page Read-Fetches-Zero Under Gapped Reads

A known scheduler/CAM interaction: under open-page policy with gapped read traffic, a read can be fetched before its data is valid (returns zero) in a narrow timing window (reproduces at `PUMICE_SEED=2`, the burst-pause / hit-miss-oscillation pattern). **Root-cause and fix** the read-fetch gating in the open-page path before promoting the HAS to formal status. The `r_fdone` fill-complete gating in the CAMs is the relevant mechanism.

30.2 TODOs Before v0.2

- Add bit-level pinout tables for the AXI, DFI, and cpuif (register) interfaces
- Add timing diagrams for the WR / RD command issue, including CWL / CL alignment
- Add sequence diagrams for `init_sequencer` (DDR2 and LPDDR2 MR init) and, where present, the optional `powerdown_ctrl`; document the FSM-free `bank_timer` countdown timers rather than an FSM diagram
- Add a draft of the Verilog package skeletons for `ddr2_init_steps_pkg.sv` and `lpddr2_init_steps_pkg.sv`
- Cross-reference each section to the corresponding pre-aspec.md bullet
- Add quantitative area / power estimates per module (synthesis pass needed)
- Add waveform examples for the canonical init sequences

30.3 Feature Roadmap — planned advanced modes (ordered)

The full catalog of planned, config-bit-selectable advanced modes lives in `docs/design-requirements.md` (“Advanced modes — selectable scheduling / paging / refresh”); the family-

level split (commodity-legal here vs model-only parked for DDR3/DDR4) is in `projects/components/memory-controllers/ADVANCED_MODES_ROADMAP.md`. Entry gate (TASKS.md TASK-FEATURES): board reads validated at the bring-up tuple, refresh-collision fix re-soaked on silicon.

Serial pre-silicon implementation order (each step OFF-by-default with its own red-to-green model test):

1. **Foundation** — `SCHED_POLICY` / `PAGE_POLICY_CFG` / `REFRESH_MODE` mode-select CSRs + `*_STATS` telemetry + PHY capability straps (no behavior change; defaults bit-identical).
2. **Scheduling (Axis 1)** — `in_order` -> `fr_fcfs` (confirm current) -> `age_threshold` -> `most/fewest_pending` -> `ACCESS_PREF` -> write-batching -> **QoS** (AxQOS-aware pick, `QOS_EN`).
3. **Paging (Axis 2)** — `static_open/close` (confirm) -> `fixed_open` -> `adapt_time` -> `rbl_static` -> `rbl_dyn` -> `adapt_access`.
4. **Refresh (Axis 3, commodity)** — JEDEC pull-in/postpone sweep -> `refpb_rr`.

None of the mode-select CSRs above exist in the RDL yet; open issue 13 (QoS) is subsumed by step 2 of this order.